

Mean-field based models for neutrino-nucleus cross sections

Natalie Jachowicz, Kajetan Niewczas, Alexis Nikolakopoulos, Matthias Hooft, Ashish Kumar Jha, Marco Vanderpoorten, Yari De Backer, Tania Franco-Munoz, Javier Garcia-Marcos, Raul Gonzalez-Jimenez

Overview of Ghent Theory

What the organizers first asked me to talk about

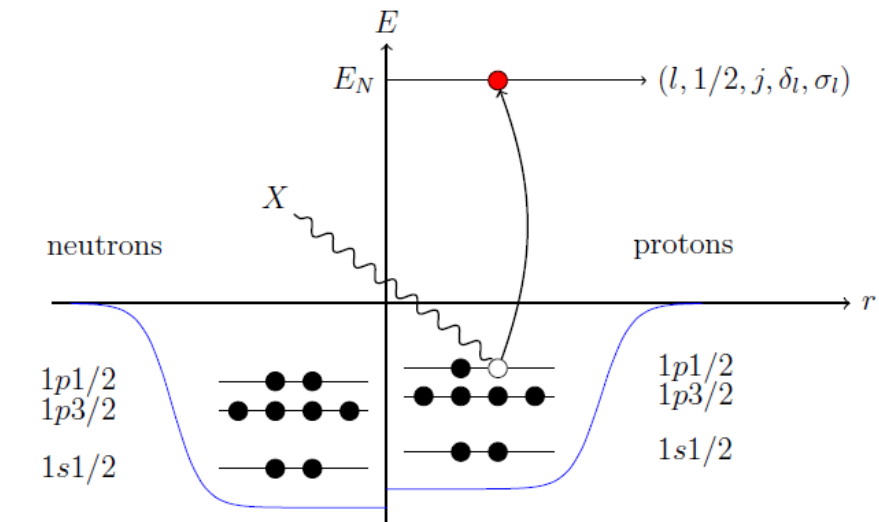
Natalie Jachowicz

Mean-field based models for neutrino-nucleus cross sections

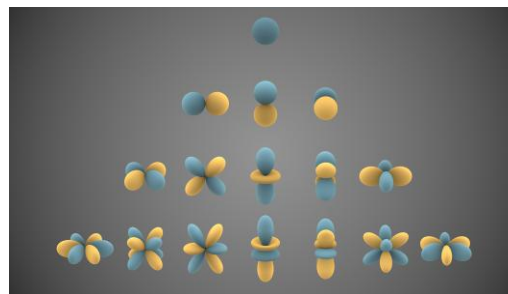
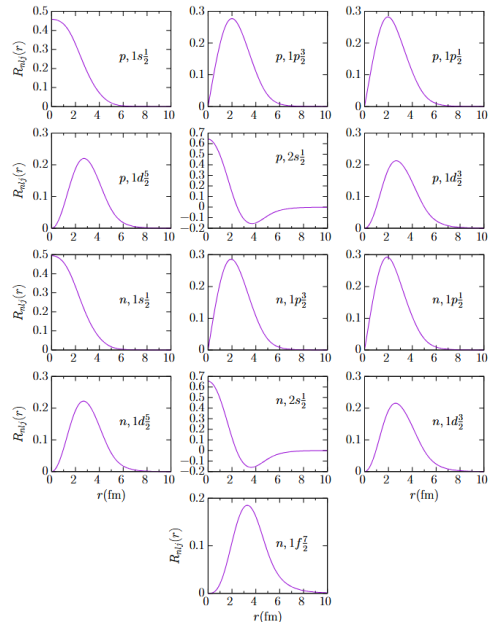
Natalie Jachowicz, Kajetan Niewczas, Alexis Nikolakopoulos, Matthias Hooft, Ashish Kumar Jha, Marco Vanderpoorten, Yari De Backer, Tania Franco-Munoz, Javier Garcia-Marcos, Raul Gonzalez-Jimenez

A Mean Field for the nucleus

- Starting point : mean-field nucleus with Hartree-Fock single-particle wave functions
- Binding energies for the different shells
- Pauli blocking fully implemented quantummechanically
- Distortion of the final nucleon : wave functions for initial bound and final nucleon calculated in the same potential
- Skyrme SkE2 force used to build the potential : effective interaction optimized to reproduce low-energy properties of closed-shell nuclei



3s	2d _{3/2}	4	
2d	3s _{1/2}	2	
	1g _{7/2}	8	
1g	2d _{5/2}	6	
	1g _{9/2}	10	50
2p	2p _{1/2}	2	
	1f _{5/2}	6	
1f	2p _{3/2}	4	
	1f _{7/2}	8	28
2s	1d _{3/2}	4	20
1d	2s _{1/2}	2	
	1d _{5/2}	6	
1p	1p _{1/2}	2	8
	1p _{3/2}	4	
1s	1s _{1/2}	2	2

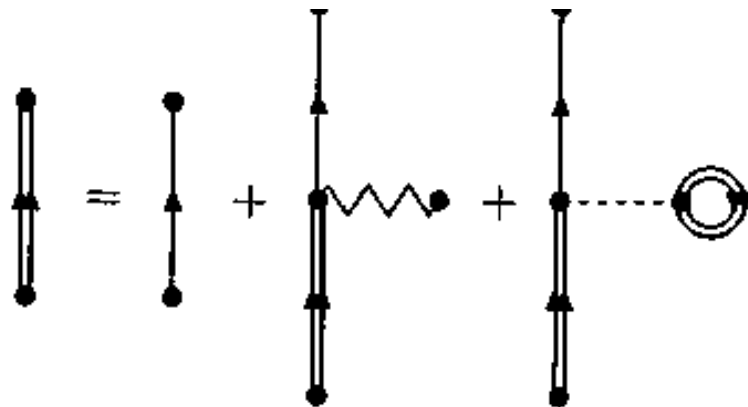


$$\begin{aligned}\hat{V}_{ij}^{(2)} = & t_0(1 + x_0 P_\sigma) \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) \\ & + \frac{1}{2} t_1 \left(\delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) k_{ij}^2 + k_{ij}'^2 \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) \right) \\ & + t_2 \mathbf{k}_{ij}' \cdot \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) \mathbf{k}_{ij} \\ & + i W_0 (\boldsymbol{\sigma}_i + \boldsymbol{\sigma}_j) \cdot (\mathbf{k}_{ij}' \times \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) \mathbf{k}_{ij}) \\ & + \frac{1}{6} t_3 (1 - x_3) (1 + P_\sigma) \rho \left(\frac{\mathbf{r}_i + \mathbf{r}_j}{2} \right) \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) \\ & + V_{ij}^{Coul}\end{aligned}$$

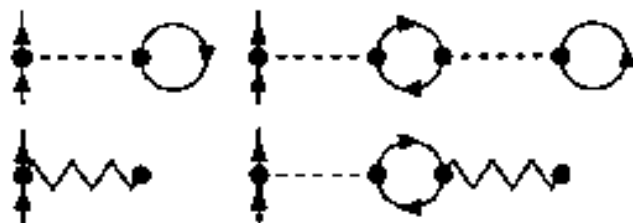
$$\begin{aligned}\hat{V}_{ijk}^{(3)} = & x_3 t_3 \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_k) \\ & + \frac{1}{6} t_4 \left[(k_{ij}'^2 + k_{jk}'^2 + k_{ki}'^2) \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_k) \cdot \right. \\ & \left. + \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_j) \delta^{(3)}(\mathbf{r}_i - \mathbf{r}_k) (k_{ij}^2 + k_{jk}^2 + k_{ki}^2) \right]\end{aligned}$$

Hartree-Fock Mean Field

$$G^{HF}(\alpha, \beta; E) = G^{(0)}(\alpha, \beta; E) + \sum_{\gamma, \delta} G^{(0)}(\alpha, \gamma; E) \Sigma^{HF}(\gamma, \delta) G^{HF}(\delta, \beta; E)$$

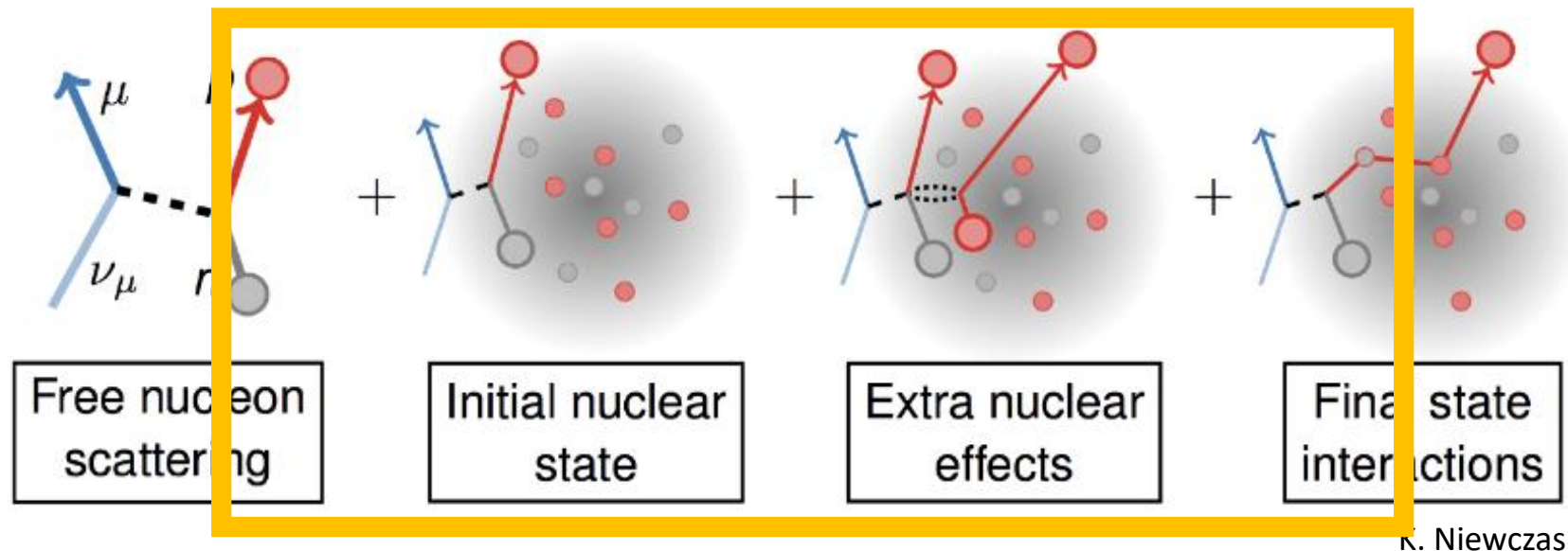


$$\Sigma^{HF}(\gamma, \delta; E) = -\langle \gamma | U | \delta \rangle - i \int \frac{dE'}{2\pi} \sum_{\mu\nu} \langle \gamma\mu | V | \delta\nu \rangle G^{HF}(\nu\mu; E')$$



Mean field already contains a lot of correlations !

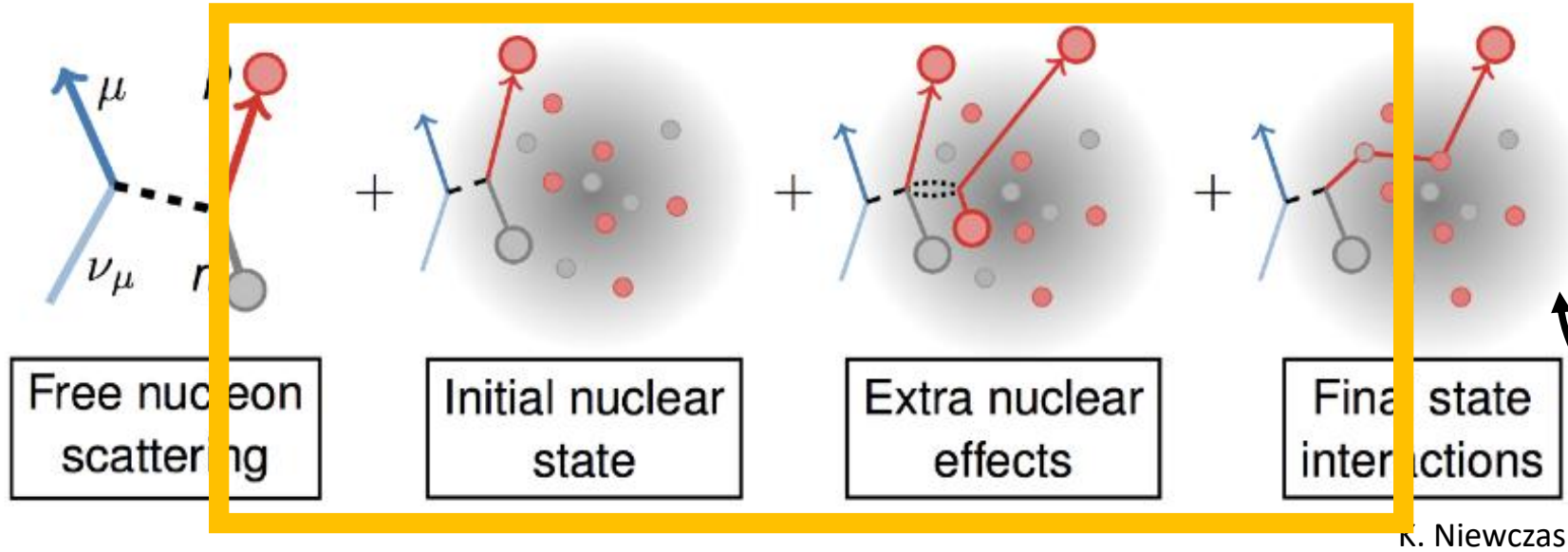
What is going on between initial and final state ... ?



- Partially taken into account by distorted waves for the final nucleon

K. Niewczas

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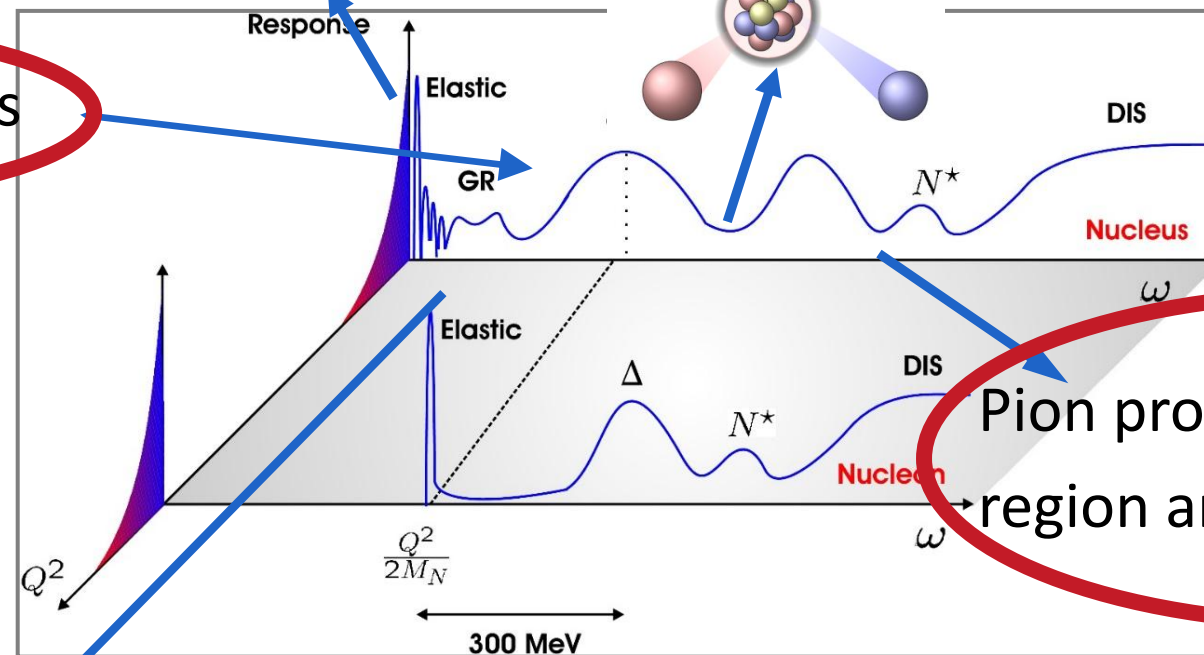
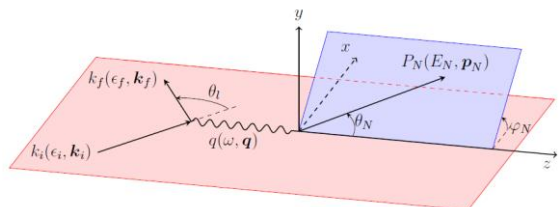


Overview

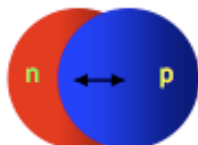
multinucleon mechanisms and 2-nucleon knockout processes in the dip region

CEvNS

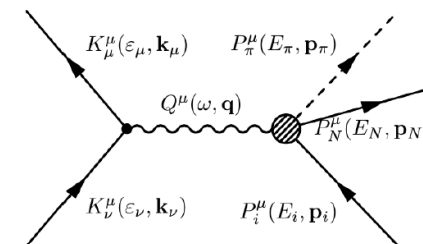
Quasi-elastic processes



Pion production in the delta region and beyond



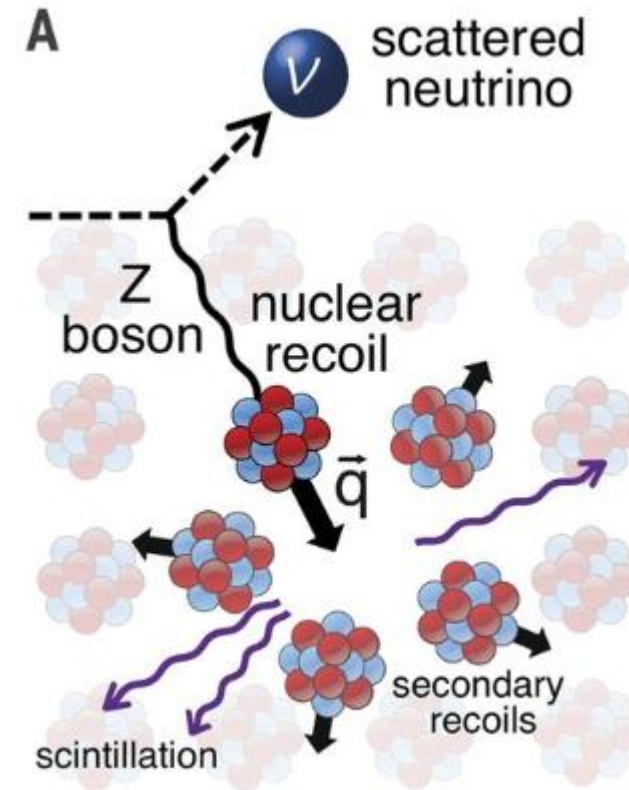
Low energy collective excitations



Coherent Scattering



Science, September 2017 : **The First Observation of Coherent Elastic Neutrino Nucleus Scattering**



Coherent Scattering

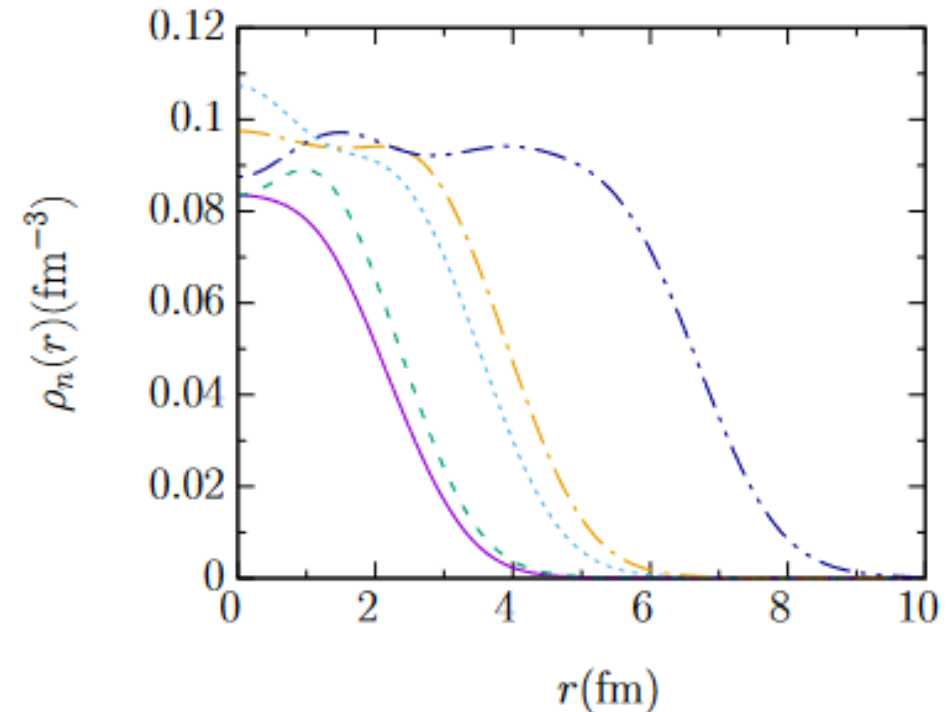
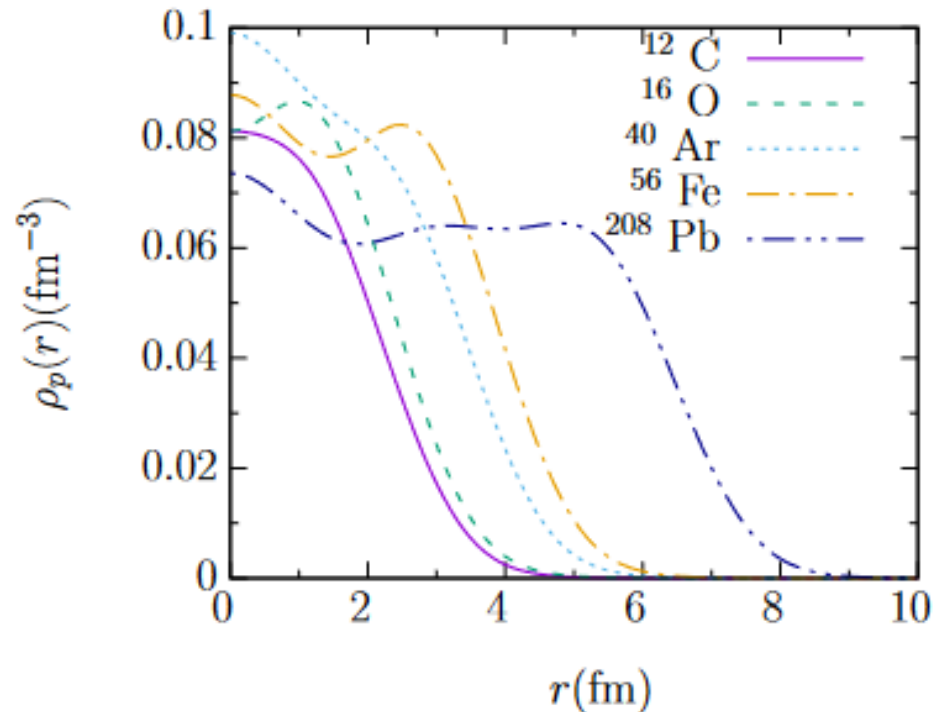
Coherent cross section as a function of nuclear recoil energy :

$$\frac{d\sigma}{dT} = \frac{G_F^2}{\pi} M_A \left(1 - \frac{T}{E_i} - \frac{M_A T}{2E_i^2} \right) \frac{Q_W^2}{4} F_W^2(Q^2),$$

Mainly sensitive to neutron distributions

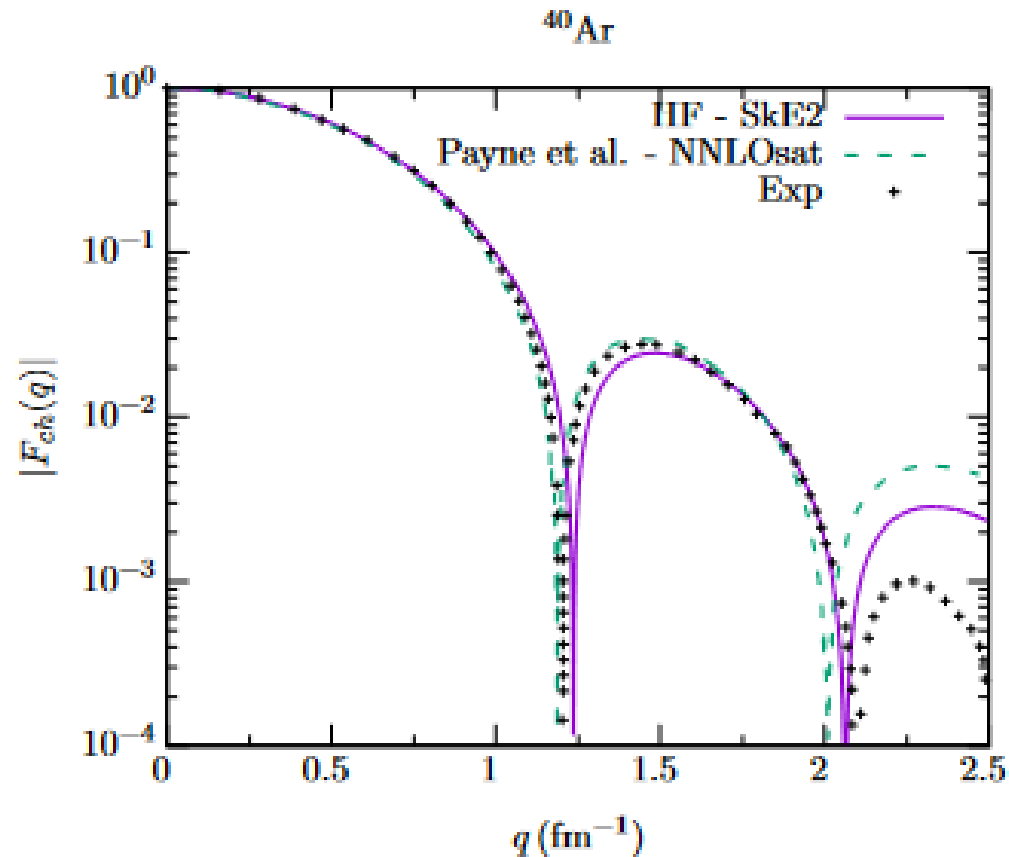
$$F(Q^2) = \frac{4\pi}{Q_W} \int \left((1 - 4 \sin^2 \theta_W) \rho_p(r) - \rho_n(r) \right) \frac{\sin(qr)}{qr} r^2 dr,$$

$$F_W(Q^2) = \frac{1}{Q_W} \left[(1 - 4 \sin^2 \theta_W) f_p(\vec{q}) F_p(Q^2) - f_n(\vec{q}) F_n(Q^2) \right]$$

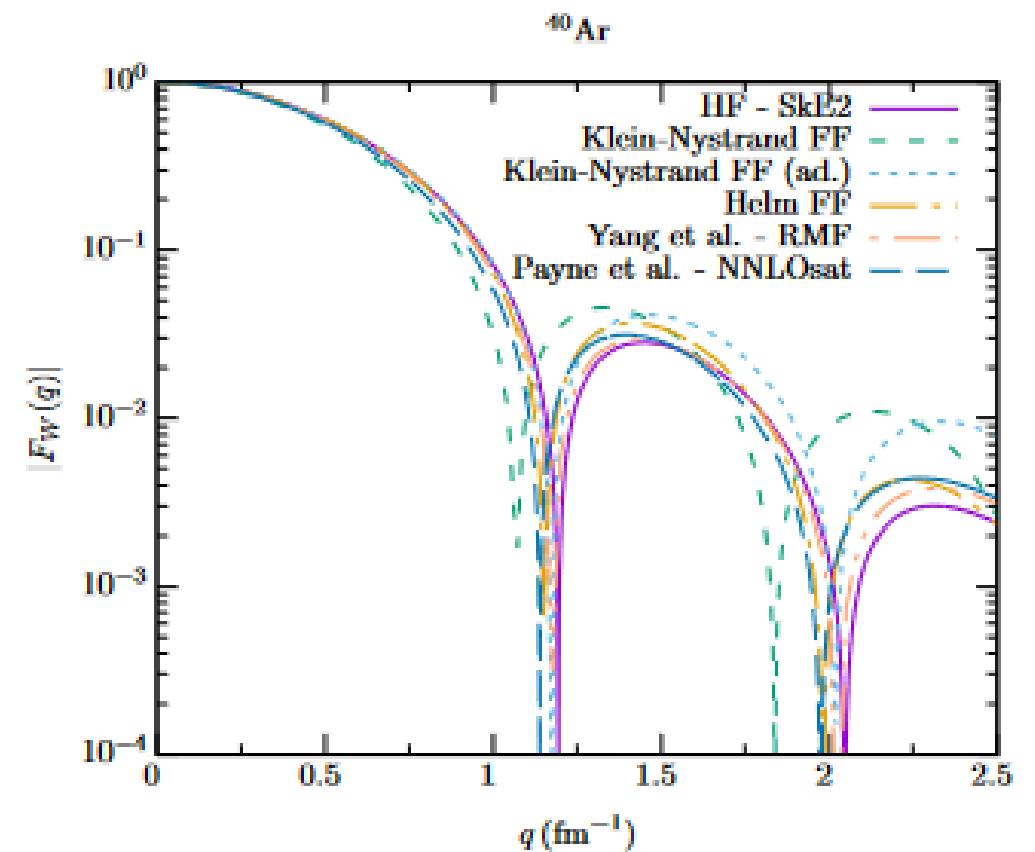


Coherent Scattering

Weak form factor : Model comparisons

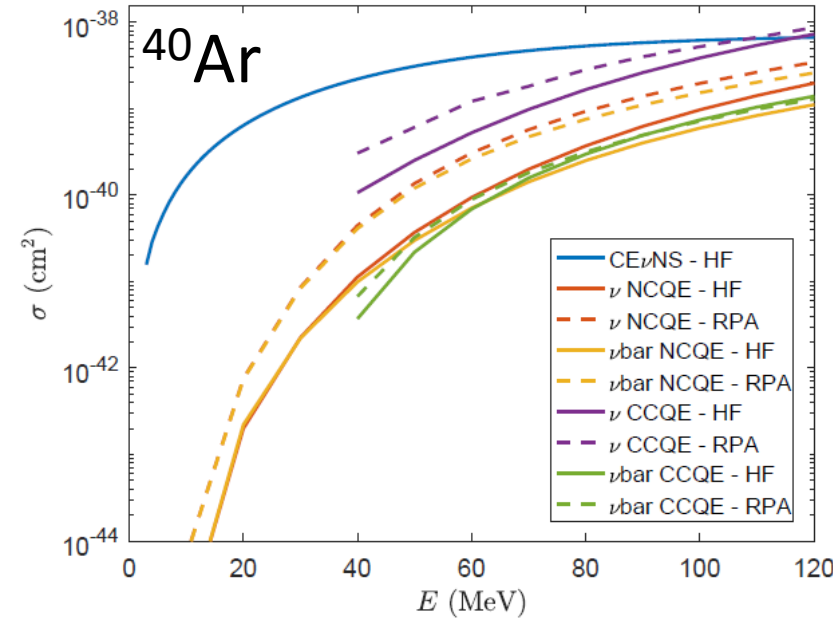
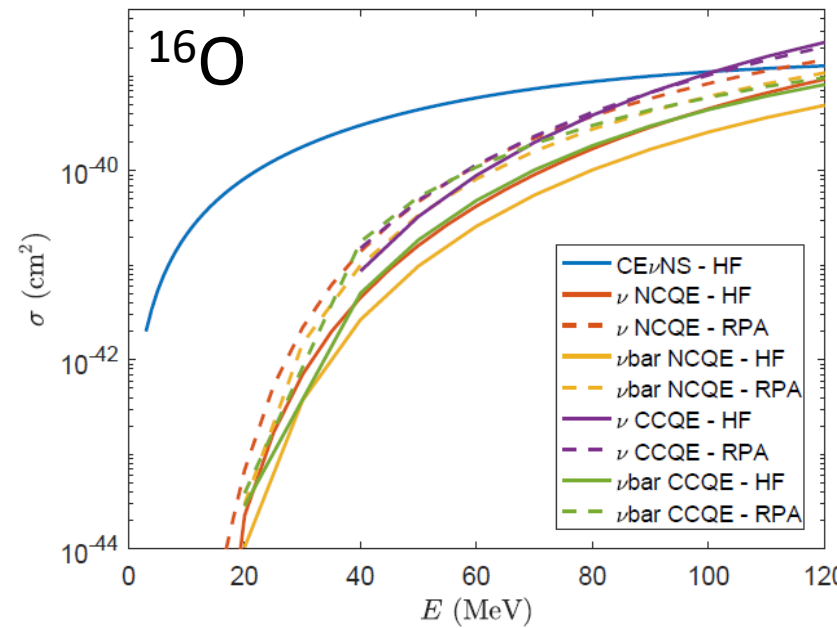
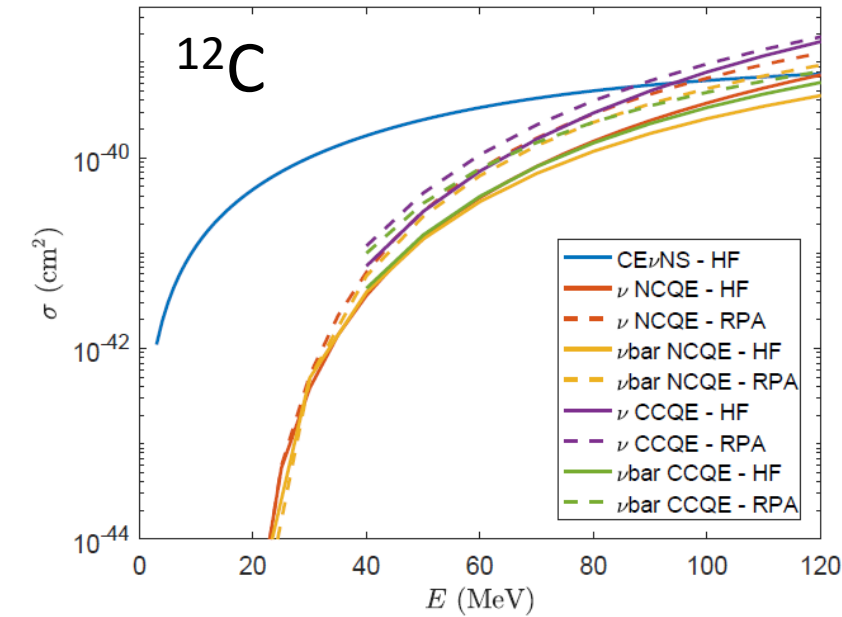


→ Robustness of the cross section against variations in description makes CevNS an ideal tool for BSM searches



N. Van Dessel, V. Pandey, H. Ray, NJ., Universe 9, 207 (2023)

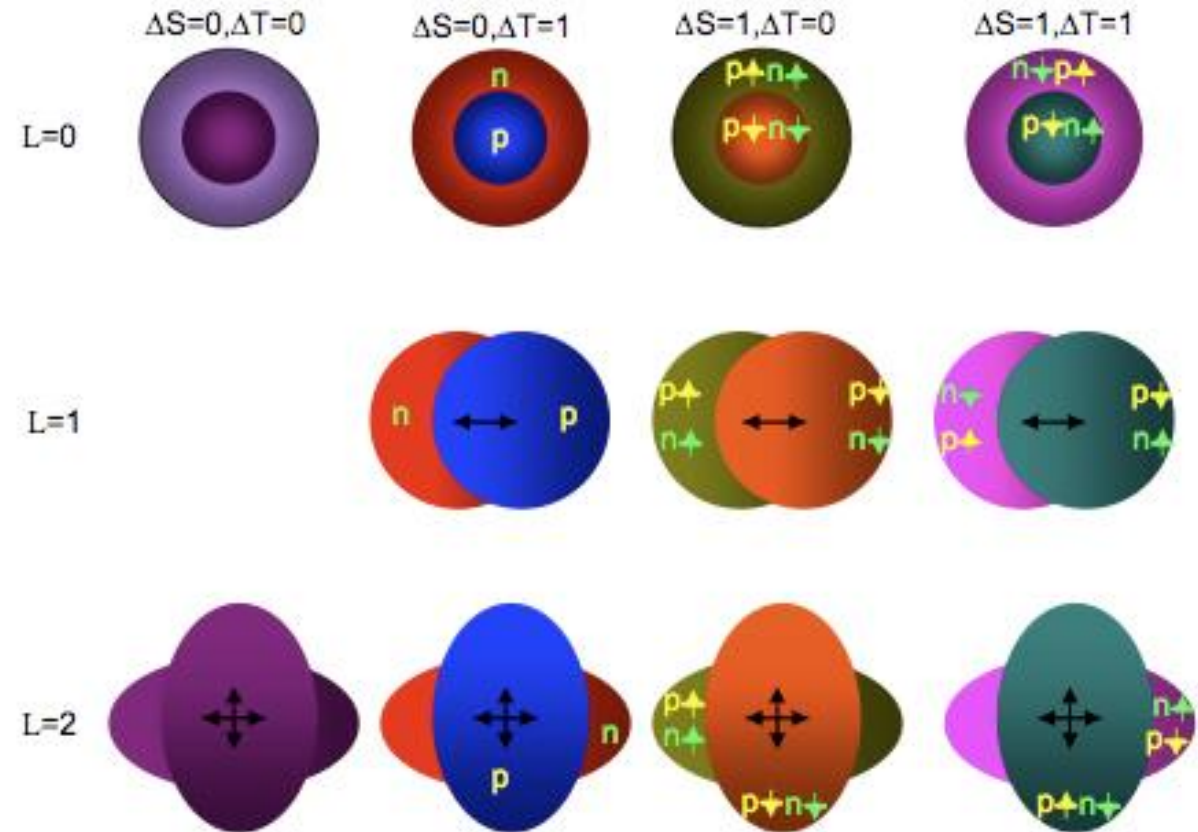
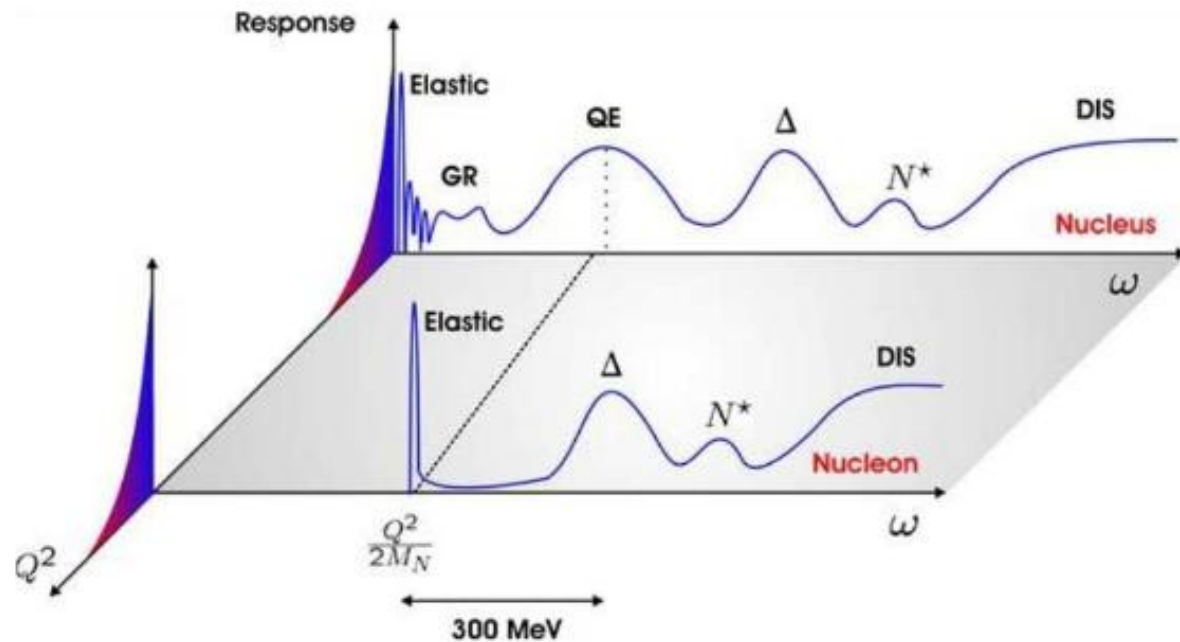
Coherent Scattering



- Strong mass dependence of coherent cross section
- Coherent process stronger than inelastic over a large kinematic range

QE-ish processes : long-range RPA correlations

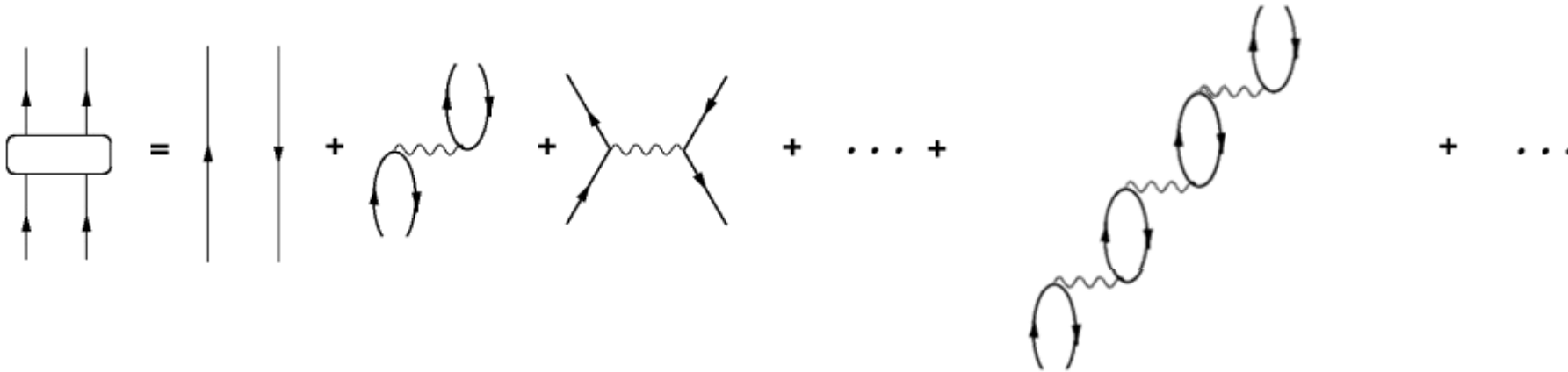
- Correlations over the whole size of the nucleus
- Correction to mean-field ground state
- Redistribute the incoming energy transfer to the nucleus over all the nuclear constituents
- Manifest themselves in collective excitations such as giant resonances at ~ 10 s of MeVs excitation energy
- They reduce strength at higher energies



<https://cyclotron.tamu.edu/research/nuclear-structure/>

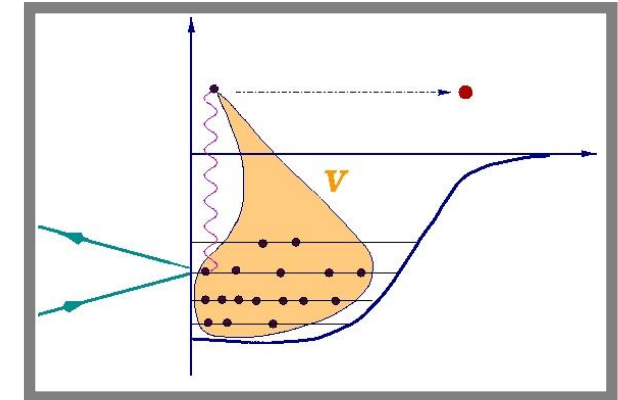
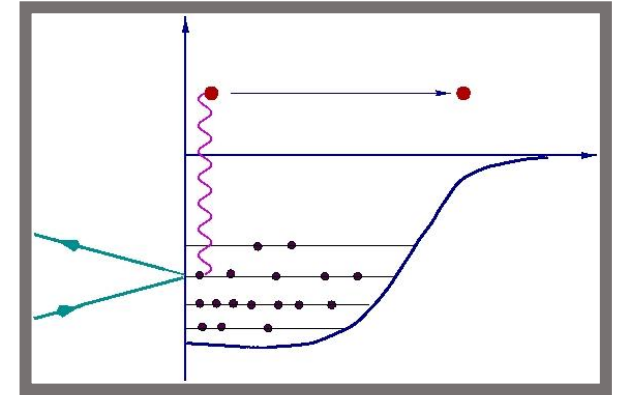
Hartree Fock mean field + Continuum RPA long-range correlations

- Green's function approach
- Skyrme SkE2 residual interaction : effective interaction optimized to reproduce low-energy properties of closed-shell nuclei
- Self-consistent calculations : same interaction is used to build the mean-field potential as to generate the long-range correlations



$$|\Psi_{RPA}\rangle = \sum_c \left\{ X_{(\Psi,C)} |ph^{-1}\rangle - Y_{(\Psi,C)} |hp^{-1}\rangle \right\}$$

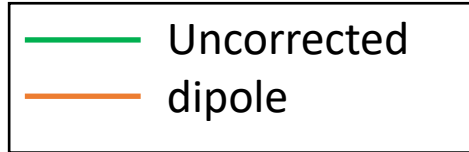
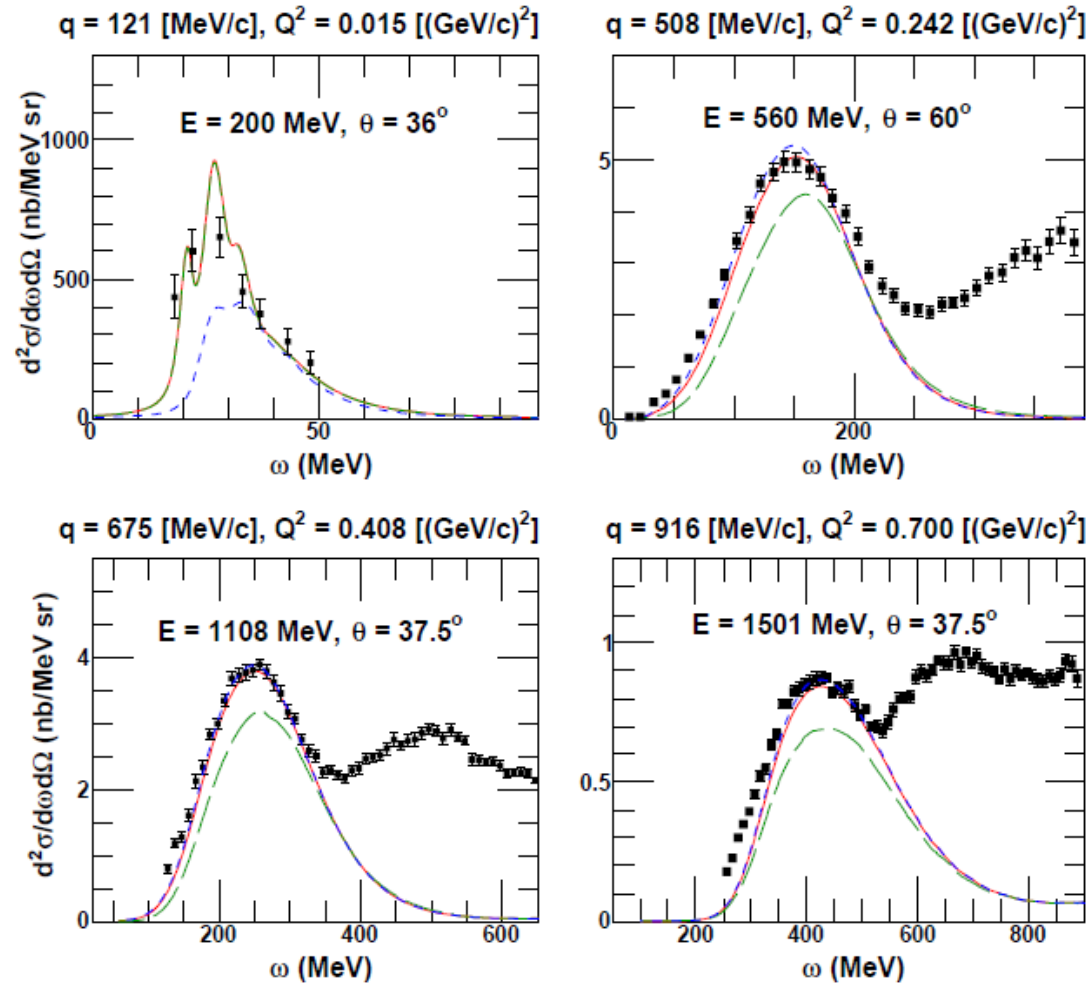
RPA wave functions for ground state and excited states : excited states described as combinations of particle-hole and hole-particle combinations out of the correlated ground state



Hartree-Fock-CRPA

- Regularization of the residual interaction :

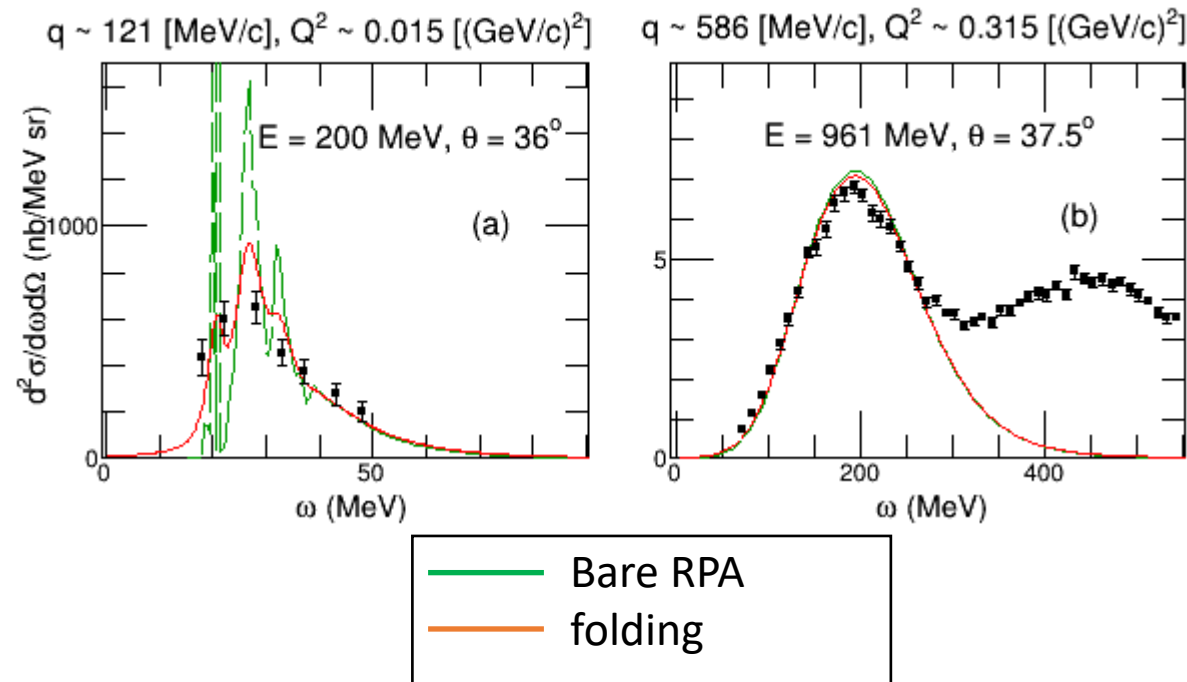
$$V(Q^2) \rightarrow V(Q^2 = 0) \frac{1}{(1 + \frac{Q^2}{\Lambda^2})^2}$$



Hartree-Fock-CRPA

Final state interactions :

- taken into account through the calculations of the wave function of the outgoing nucleon in the (real) nuclear potential generated using the Skyrme force
- influence of the **spreading width of the particle states** is implemented through a folding procedure



$$R'(q, \omega') = \int_{-\infty}^{\infty} d\omega R(q, \omega) L(\omega, \omega'),$$

$$L(\omega, \omega') = \frac{1}{2\pi} \left[\frac{\Gamma}{(\omega - \omega')^2 + (\Gamma/2)^2} \right].$$

Hartree-Fock-CRPA

•Coulomb correction for the outgoing lepton in charged-current interactions :

- ✓ Low energies : Fermi function

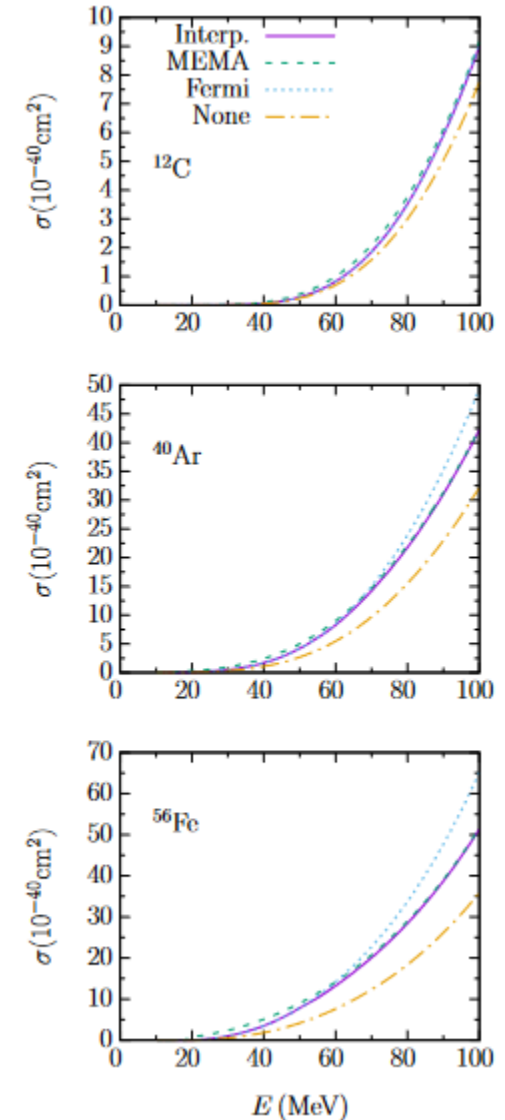
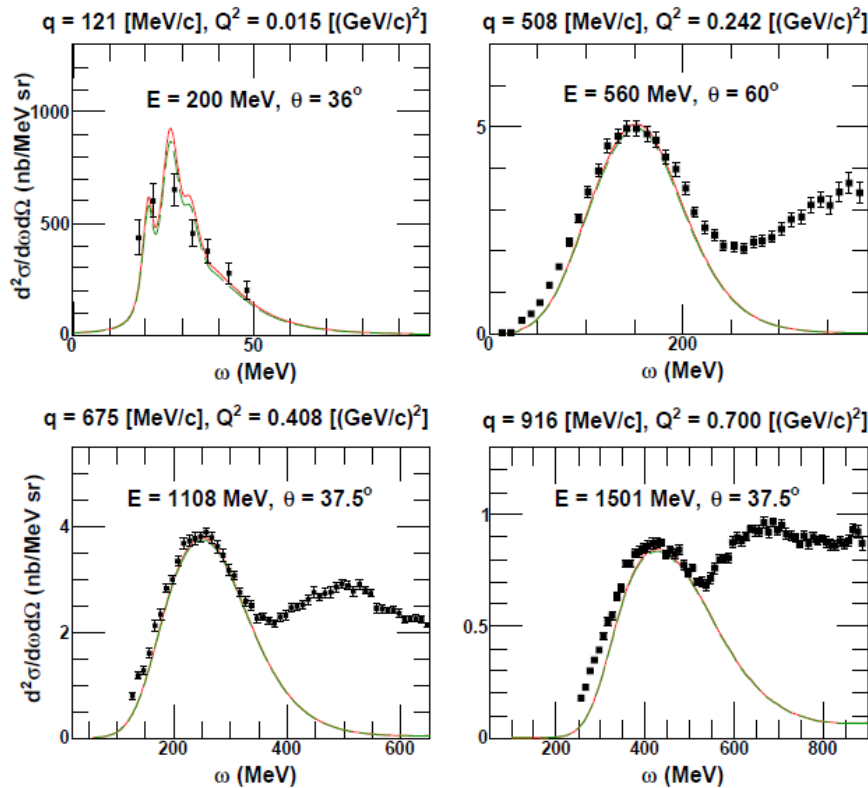
$$F(Z', E) = \frac{2\pi\eta}{1 - e^{-2\pi\eta}} \quad \eta \sim \mp Z'\alpha$$

- ✓ Higher energies : modified effective momentum approximation (J. Engel, PRC57 2004)

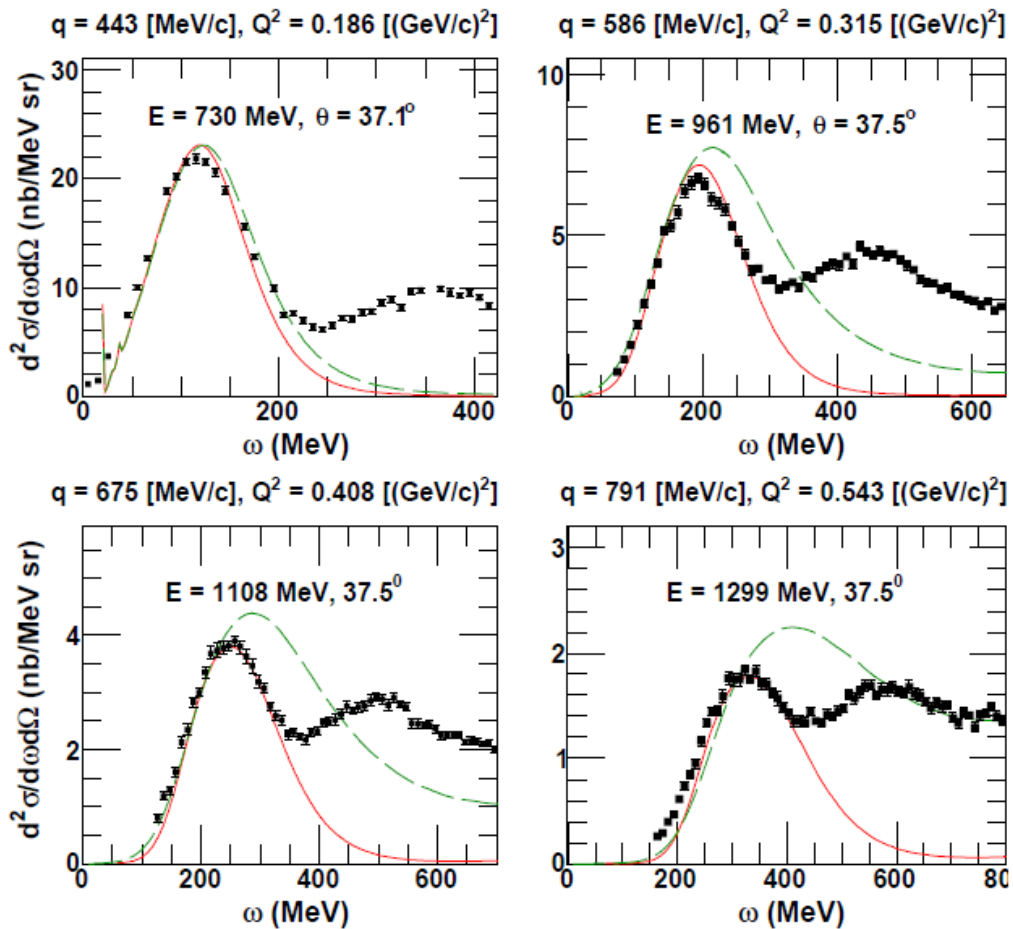
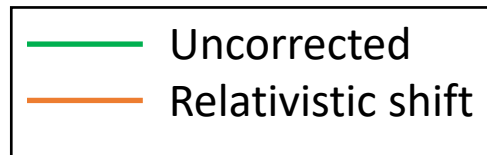
$$q_{eff} = q + 1.5 \left(\frac{Z'\alpha\hbar c}{R} \right),$$

$$\Psi_l^{eff} = \zeta(Z', E, q) \Psi_l,$$

$$\zeta(Z', E, q) = \sqrt{\frac{q_{eff} E_{eff}}{qE}}$$



- **Relativistic corrections** at higher energies (S. Jeschonnek and T. Donnelly, PRC57, 2438 (1998)):



Shift in response functions :

$$\lambda \rightarrow \lambda(\lambda + 1) \quad \lambda = \omega/2M_N$$

- The outgoing nucleon obtains the correct relativistic momentum

$$p = \sqrt{T^2 + 2MT}$$
- Shifts the QE peak to the right relativistic position

Boost :

$$R_{CC}^V(q, \omega) \rightarrow \frac{q^2}{q^2 - \omega^2} R_{CC}^V(q, \omega),$$

$$R_{LL}^A(q, \omega) \rightarrow \left(1 + \frac{q^2 - \omega^2}{4m^2}\right) R_{LL}^A(q, \omega),$$

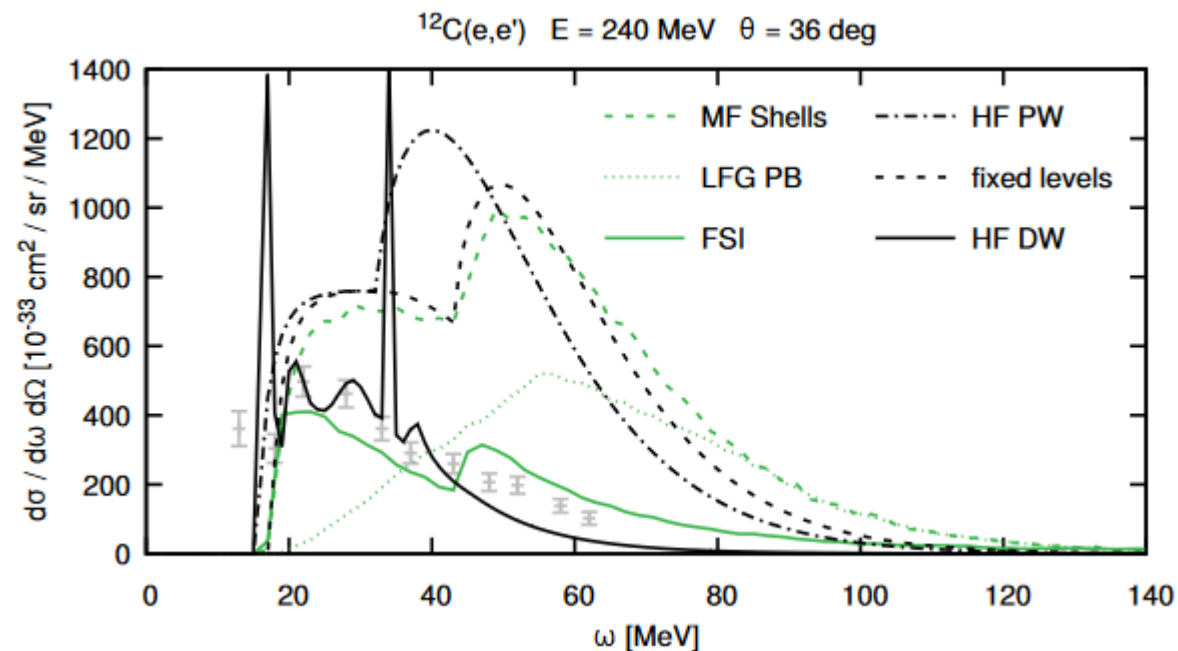
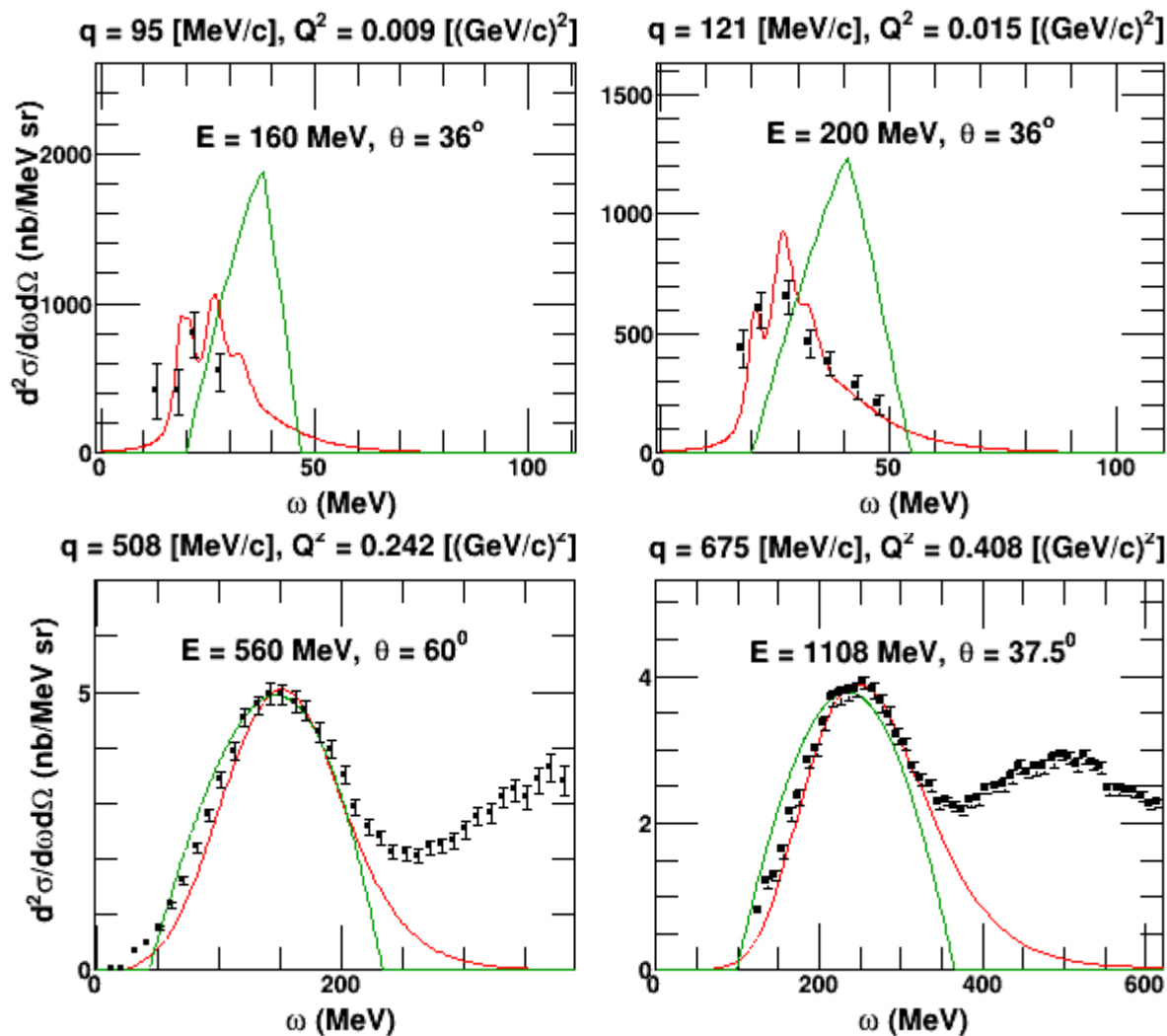
$$R_T^V(q, \omega) \rightarrow \frac{q^2 - \omega^2}{q^2} R_T^V(q, \omega),$$

$$R_T^A(q, \omega) \rightarrow \left(1 + \frac{q^2 - \omega^2}{4m^2}\right) R_T^A(q, \omega),$$

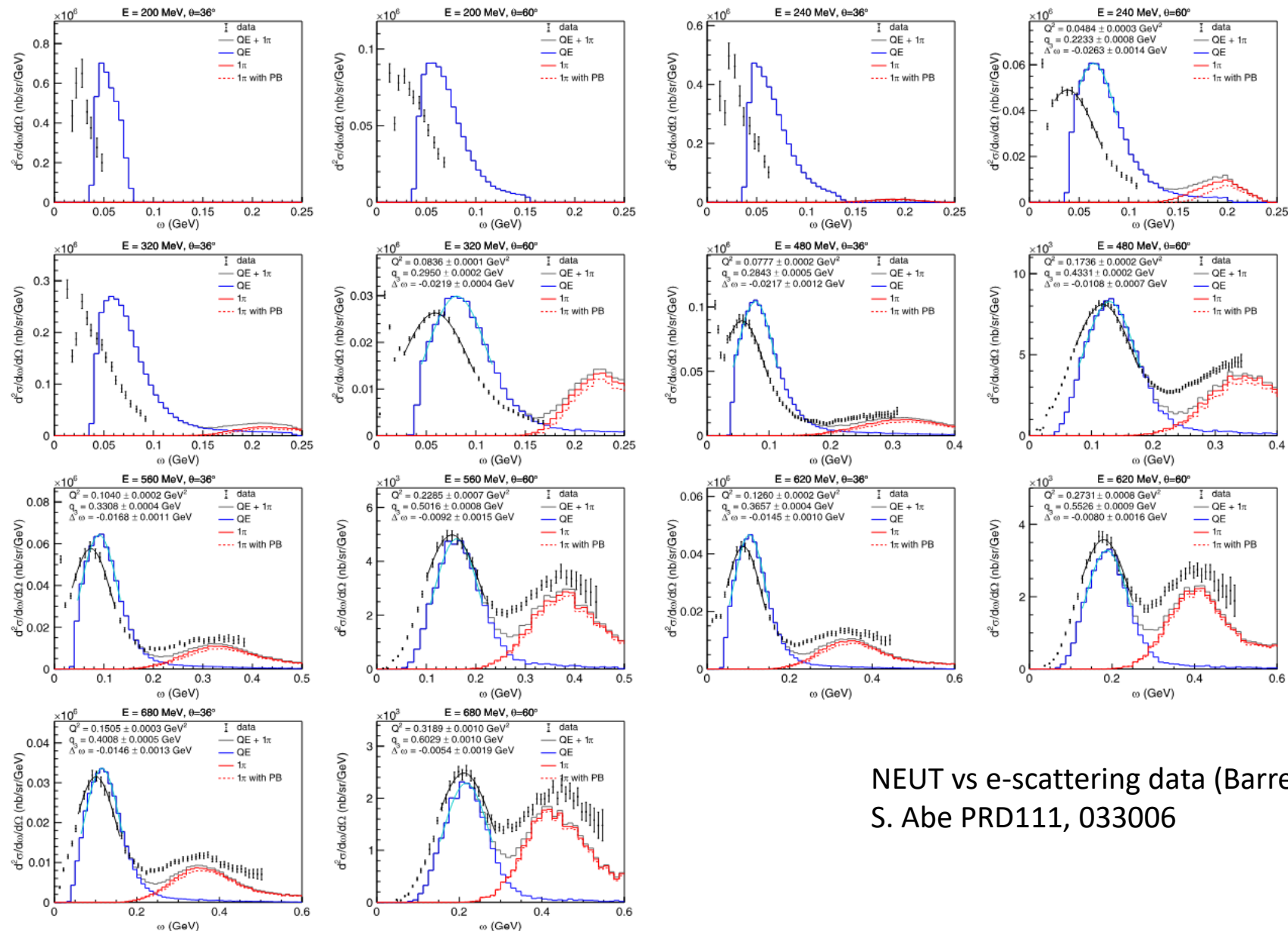
$$R_{T'}^{VA}(q, \omega) \rightarrow \sqrt{\frac{q^2 - \omega^2}{q^2}} \sqrt{1 + \frac{q^2 - \omega^2}{4m^2}} R_{T'}^{VA}(q, \omega).$$

Hartree-Fock-CRPA

— Fermi gas
— Hartree-Fock



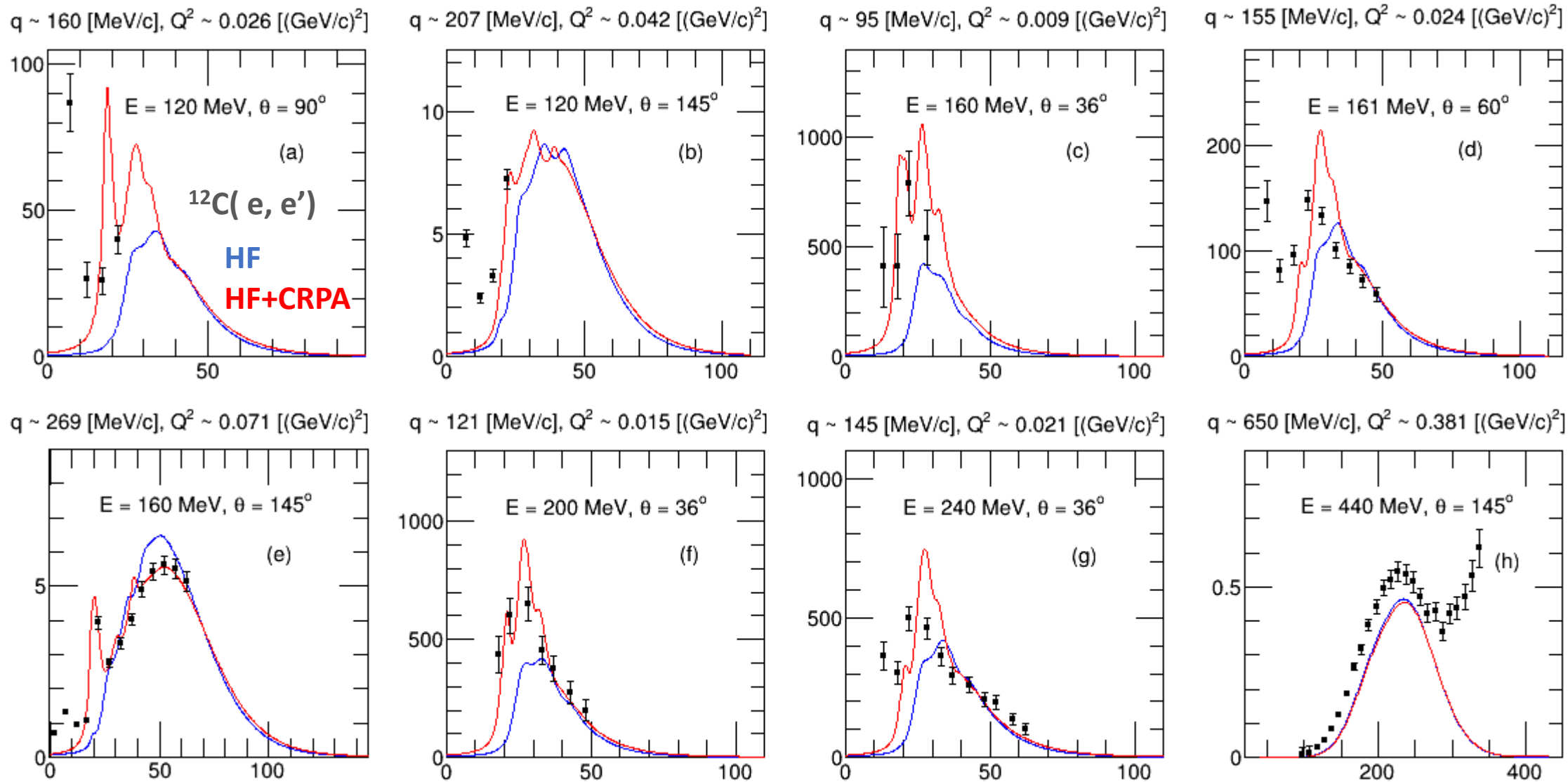
Comparison with electron scattering data



NEUT vs e-scattering data (Barreau)
S. Abe PRD111, 033006

Comparison with electron scattering data

$d^2\sigma/d\omega d\Omega(\text{nb/MeV sr})$



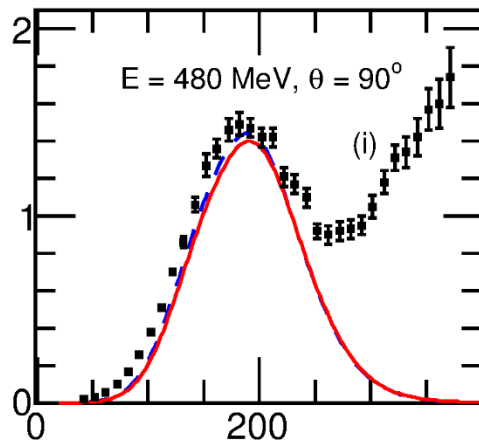
V. Pandey, N.J., J. Ryckebusch, T. Van Cuyck, M. Martini, Phys. Rev. C 92, 024606 (2015)

$\omega(\text{MeV})$

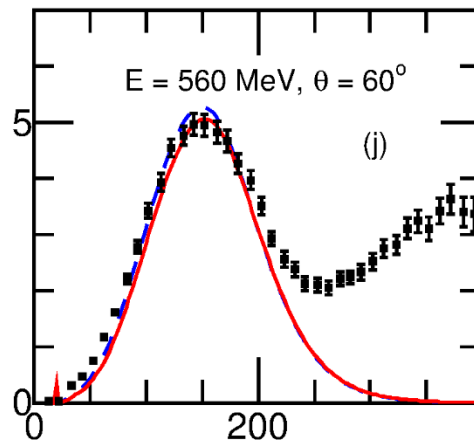
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$d^2\sigma/d\omega d\Omega(\text{nb/MeV sr})$

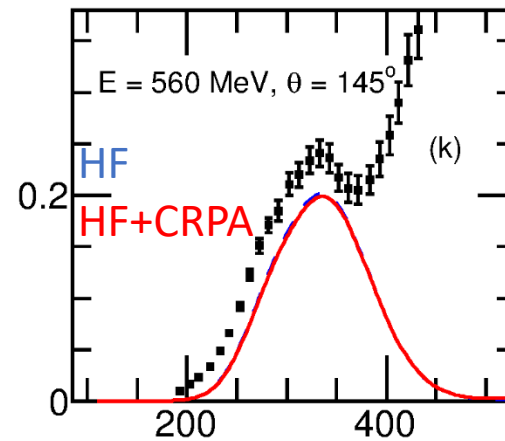
$q \sim 576 \text{ [MeV/c]}, Q^2 \sim 0.305 \text{ [(GeV/c)}^2\text{]}$



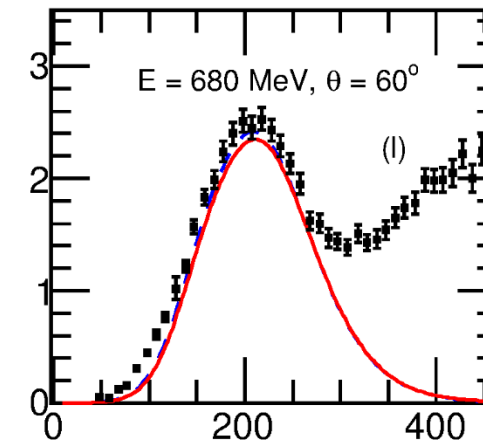
$q \sim 508 \text{ [MeV/c]}, Q^2 \sim 0.242 \text{ [(GeV/c)}^2\text{]}$



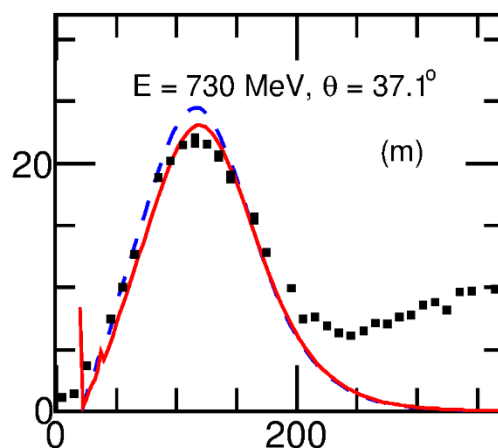
$q \sim 795 \text{ [MeV/c]}, Q^2 \sim 0.548 \text{ [(GeV/c)}^2\text{]}$



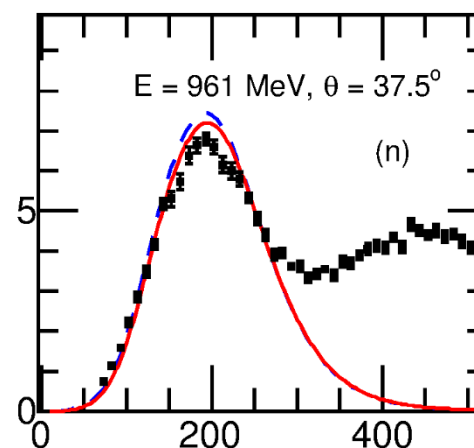
$q \sim 610 \text{ [MeV/c]}, Q^2 \sim 0.340 \text{ [(GeV/c)}^2\text{]}$



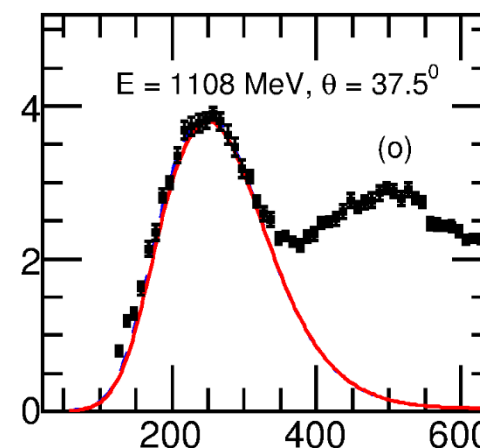
$q \sim 443 \text{ [MeV/c]}, Q^2 \sim 0.186 \text{ [(GeV/c)}^2\text{]}$



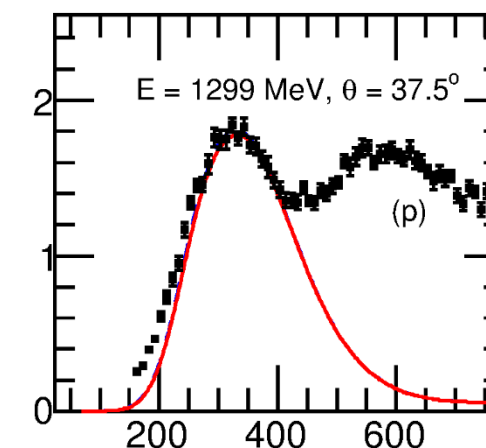
$q \sim 586 \text{ [MeV/c]}, Q^2 \sim 0.315 \text{ [(GeV/c)}^2\text{]}$



$q \sim 675 \text{ [MeV/c]}, Q^2 \sim 0.408 \text{ [(GeV/c)}^2\text{]}$



$q \sim 791 \text{ [MeV/c]}, Q^2 \sim 0.543 \text{ [(GeV/c)}^2\text{]}$

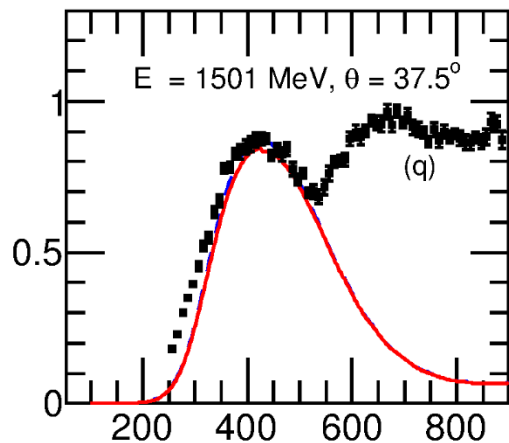


V. Pandey, N.J., J. Ryckebusch, T. Van Cuyck, M. Martini, Phys. Rev. C 92, 024606 (2015) $\omega \text{ (MeV)}$

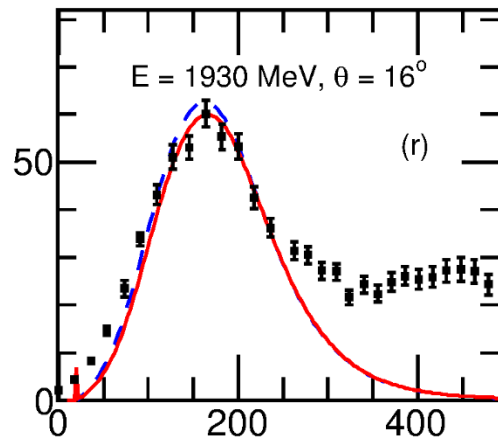
Comparison with electron scattering data

$d^2\sigma/d\omega d\Omega(\text{nb/MeV sr})$

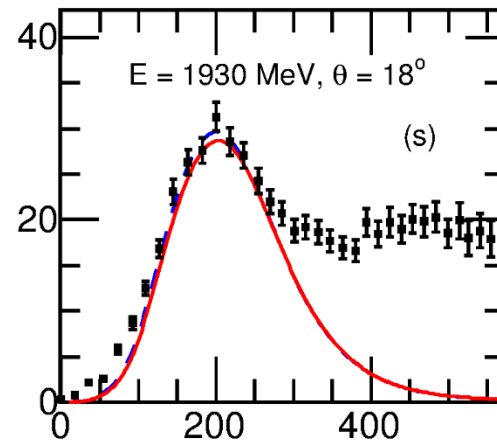
$q \sim 916 \text{ [MeV/c]}, Q^2 \sim 0.700 \text{ [(GeV/c)}^2\text{]}$



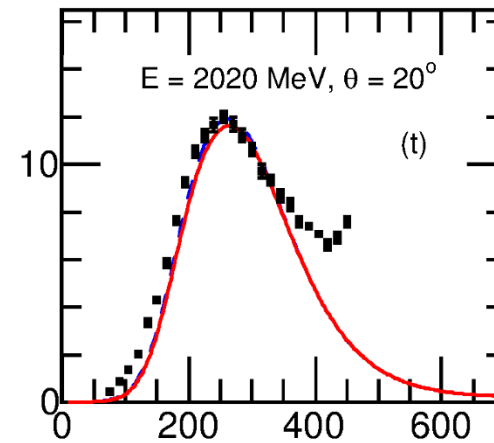
$q \sim 536 \text{ [MeV/c]}, Q^2 \sim 0.267 \text{ [(GeV/c)}^2\text{]}$



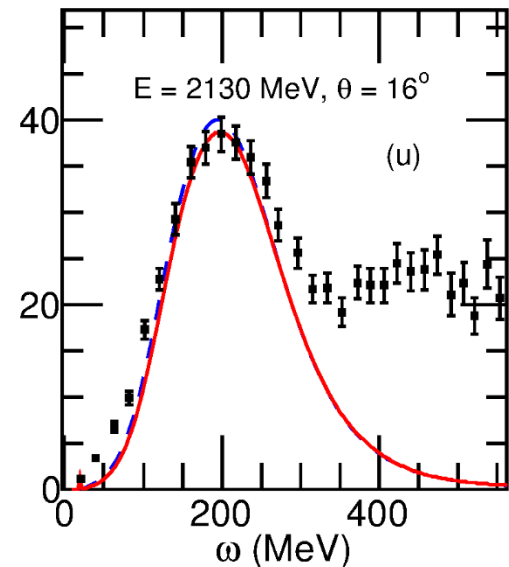
$q \sim 601 \text{ [MeV/c]}, Q^2 \sim 0.331 \text{ [(GeV/c)}^2\text{]}$



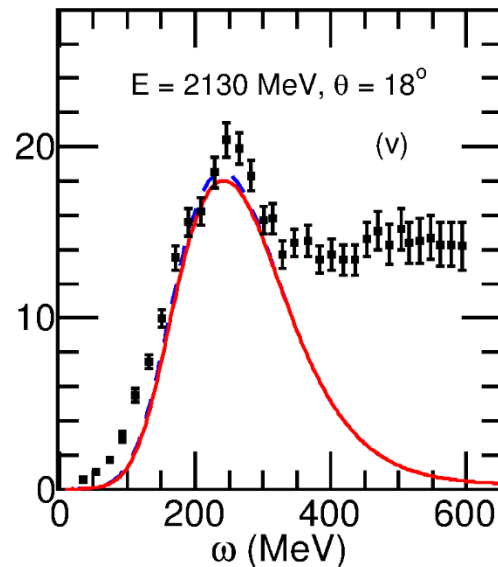
$q \sim 700 \text{ [MeV/c]}, Q^2 \sim 0.436 \text{ [(GeV/c)}^2\text{]}$



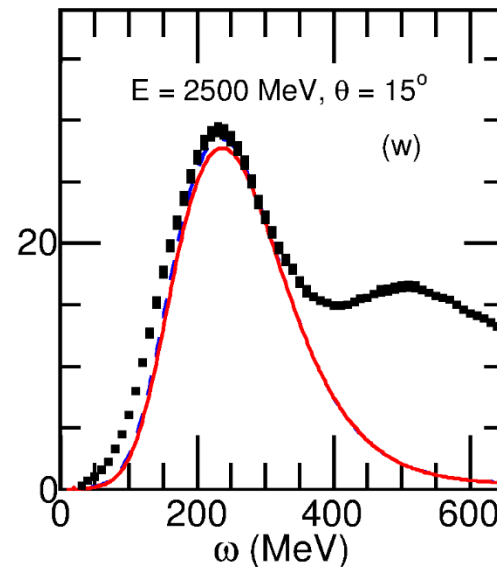
$q \sim 594 \text{ [MeV/c]}, Q^2 \sim 0.323 \text{ [(GeV/c)}^2\text{]}$



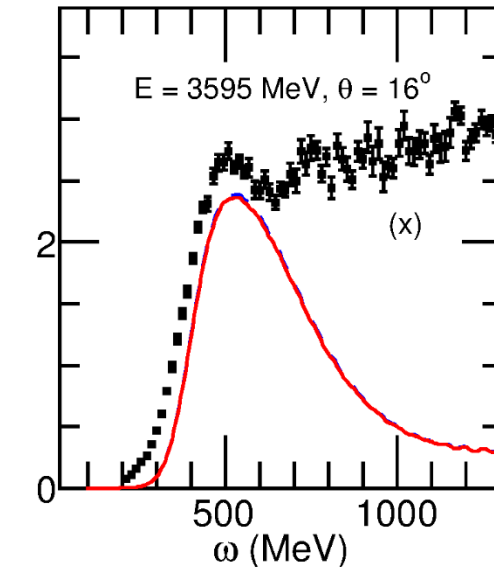
$q \sim 667 \text{ [MeV/c]}, Q^2 \sim 0.399 \text{ [(GeV/c)}^2\text{]}$



$q \sim 658 \text{ [MeV/c]}, Q^2 \sim 0.391 \text{ [(GeV/c)}^2\text{]}$

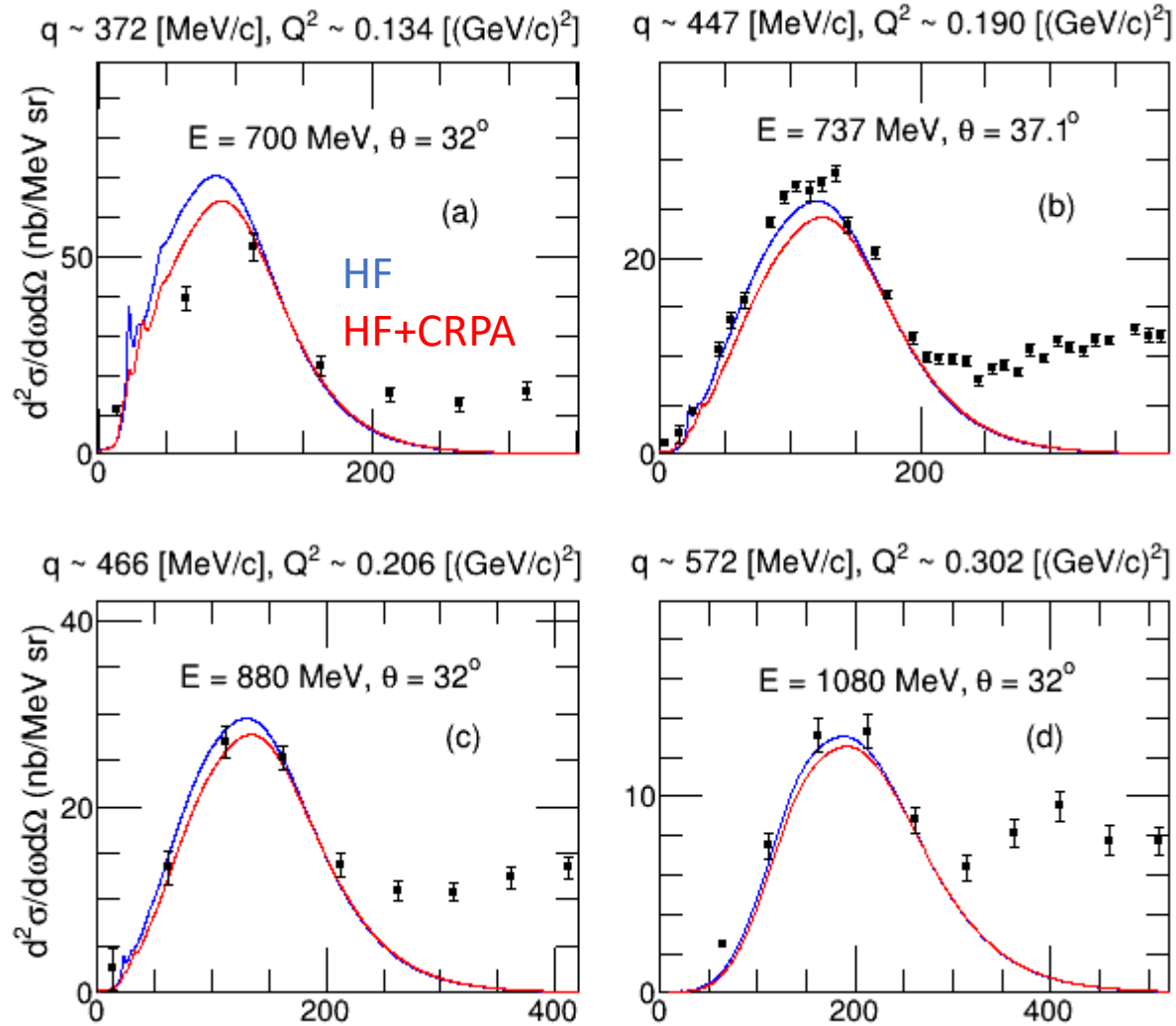


$q \sim 1043 \text{ [MeV/c]}, Q^2 \sim 0.872 \text{ [(GeV/c)}^2\text{]}$



$\omega \text{ (MeV)}$

Comparison with electron scattering data

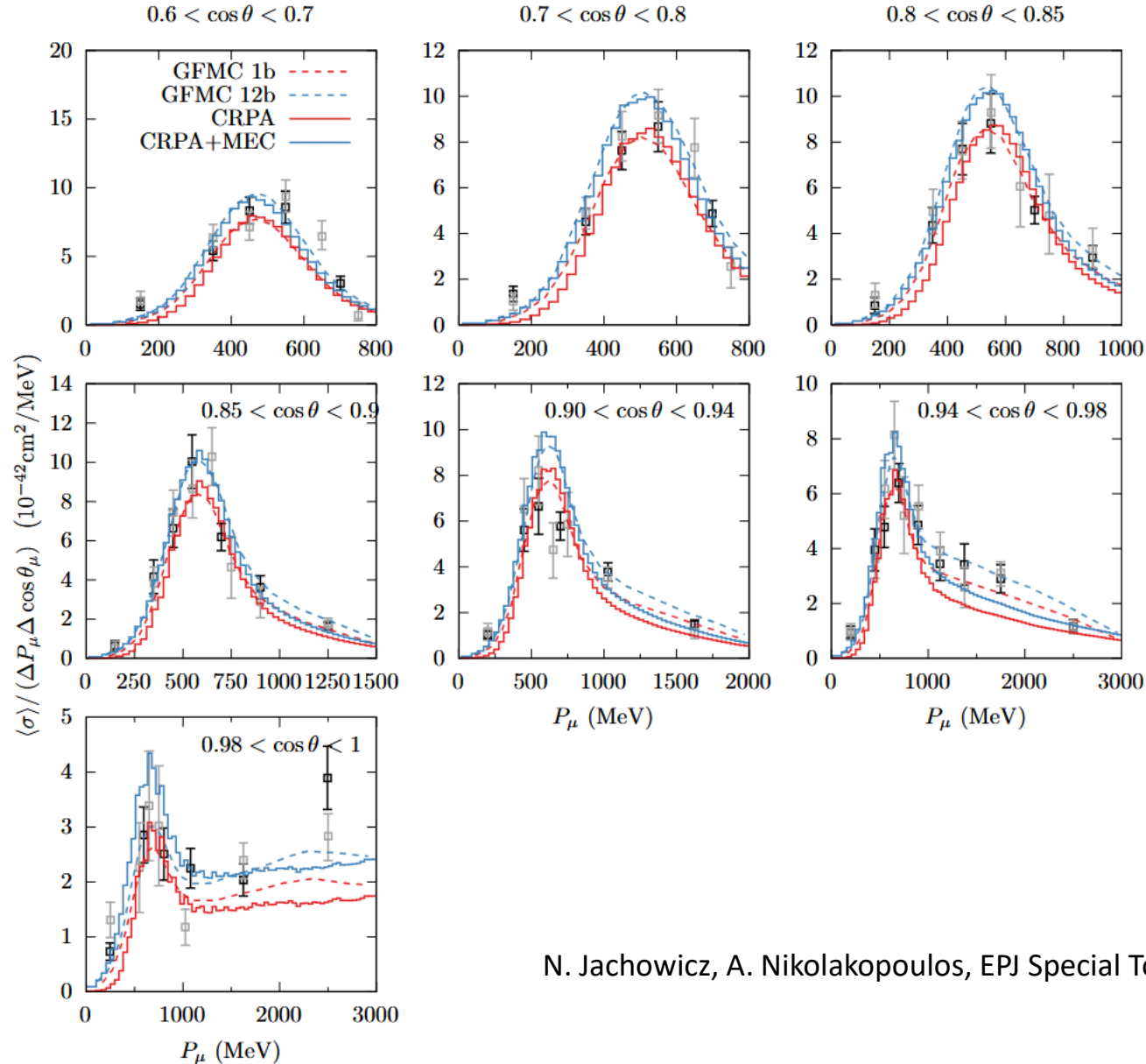


$^{16}\text{O}(e, e')$

- Good overall agreement with e-scattering data

Data : P. Barreau et al., Nucl. Phys. A402, 515 (1983), J. S. O'Connell et al., Phys. Rev. C35, 1063 (1987), R. M. Sealock et al., Phys. Rev. Lett.62, 1350 (1989), D. S. Bagdasaryan et al., YERPHI-1077-40-88 (1988), D. B. Day et al., Phys. Rev. C 48, 1849 (1993), D. Zeller, DESY-F23-73-2 (1973).

T2K flux-folded inclusive cross sections

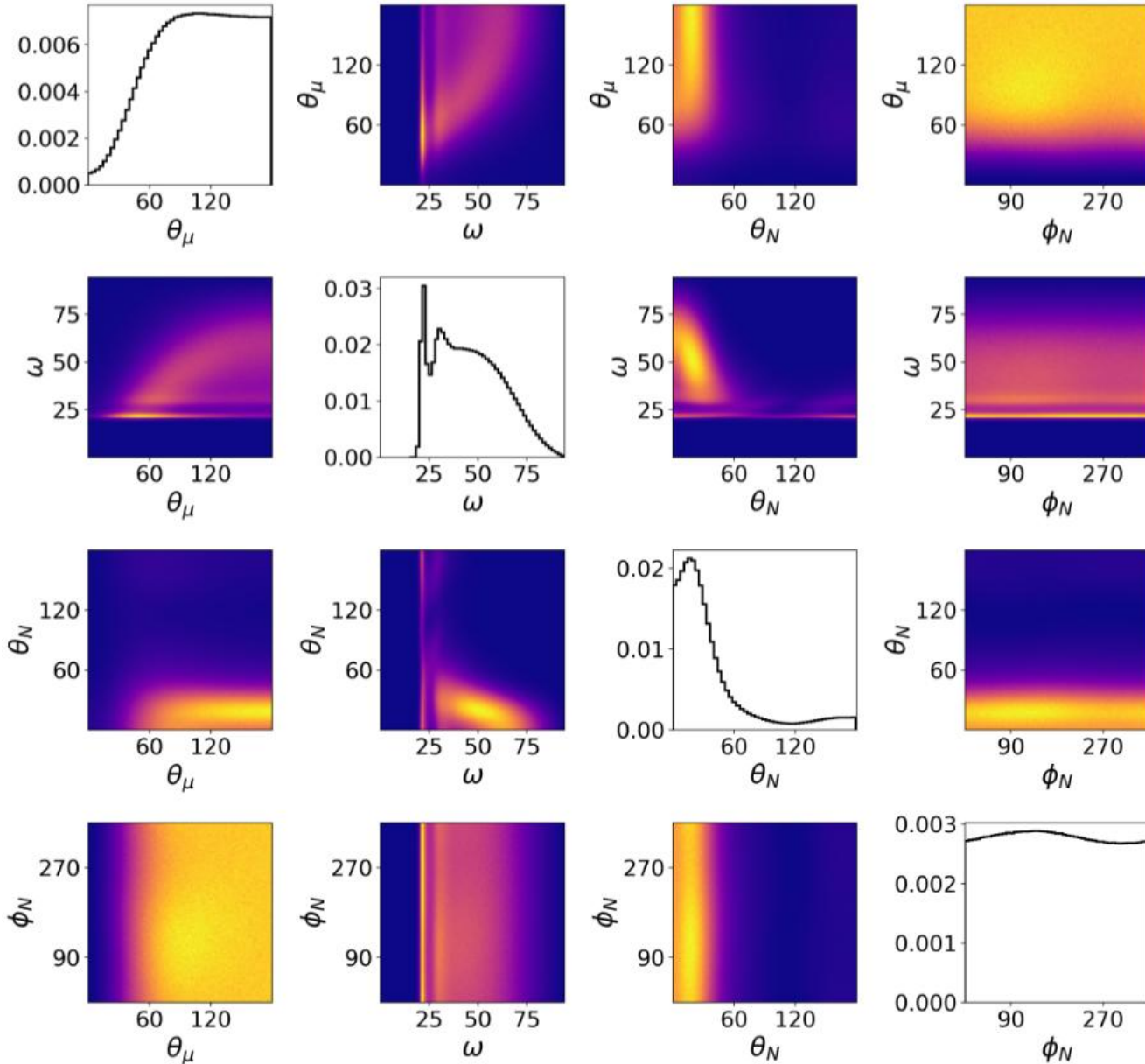


- T2K flux-folded double differential cross sections per target neutron for ν_μ -CCQE scattering on ^{12}C
- Comparing ab-initio calculations with mean-field-based calculations
- Include long-range correlations in CRPA and added (SuSA) 2p-2h contributions

N. Jachowicz, A. Nikolakopoulos, EPJ Special Topics 230, 7803

HF exclusive cross sections

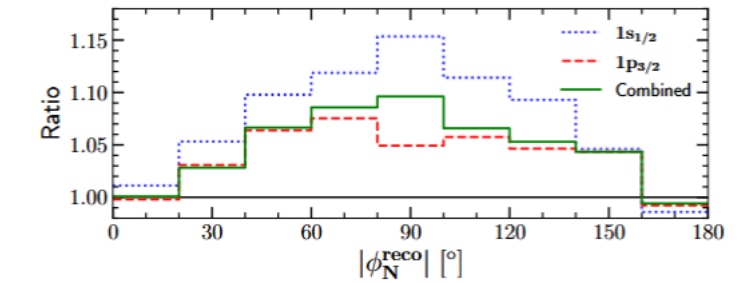
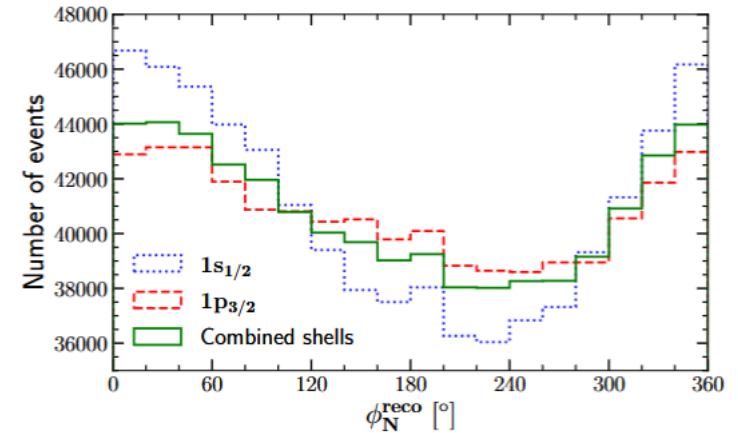
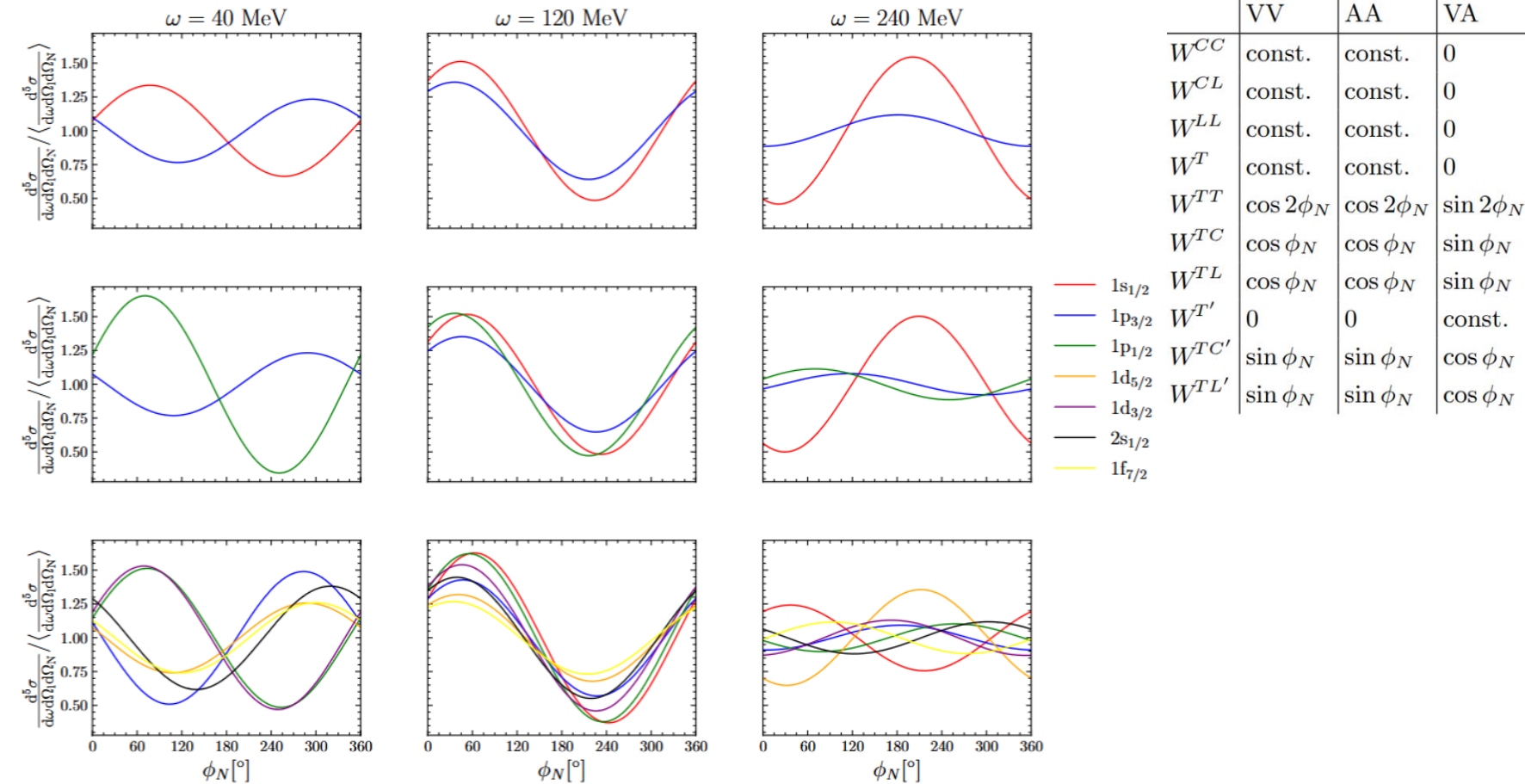
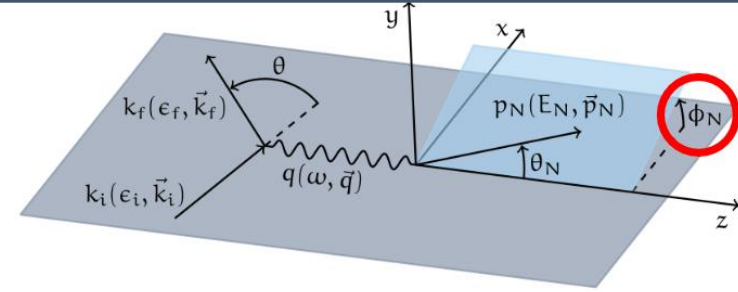
^{12}C , $E_v=200$ MeV, $p_{3/2}$ shell



M. El Baz et al.. PRD111, 1130021.

HF exclusive cross sections : azimuthal dependence

- Interplay of parity violation and distortion of outgoing nucleon wave by the potential



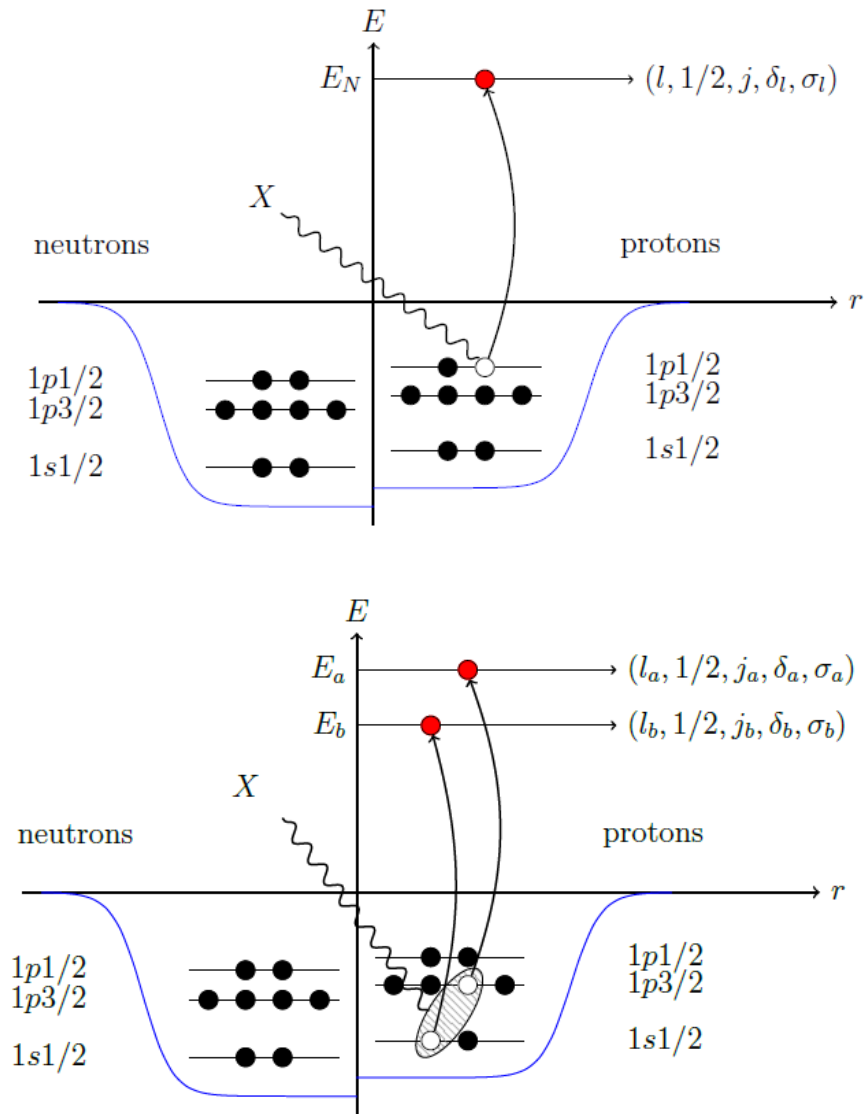
M. Vanderpoorten et al., arXiv2605.26877.

Multinucleon mechanisms

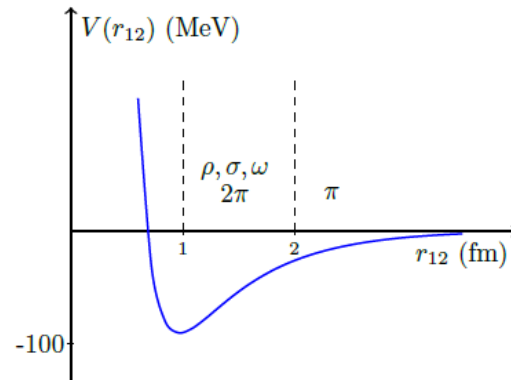
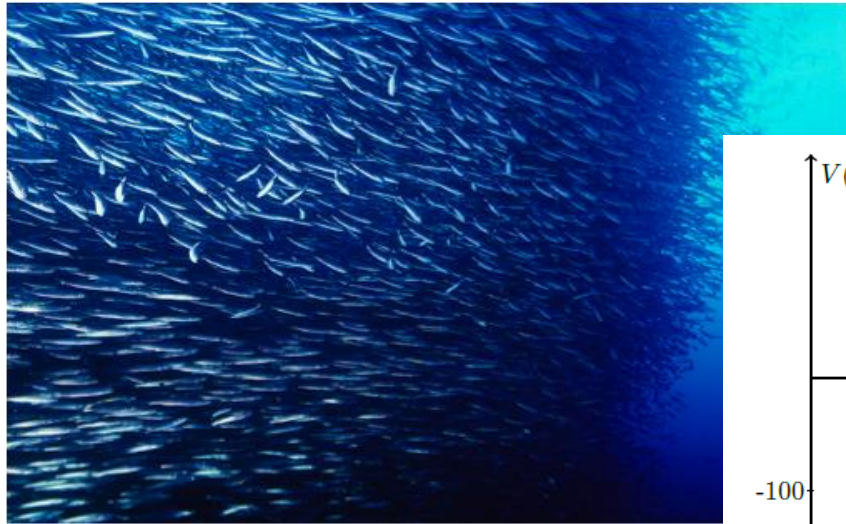
Cross section calculations

Nuclear model

- Starting point : mean-field nucleus with **Hartree-Fock** single-particle wave functions
- Skyrme SkE2 force used to build the potential
- Pauli blocking
- Binding
- Long-range correlations
- **Short-range correlations**
- **Meson Exchange Currents**
- **2p2h processes and influence on 1p1h cross sections**



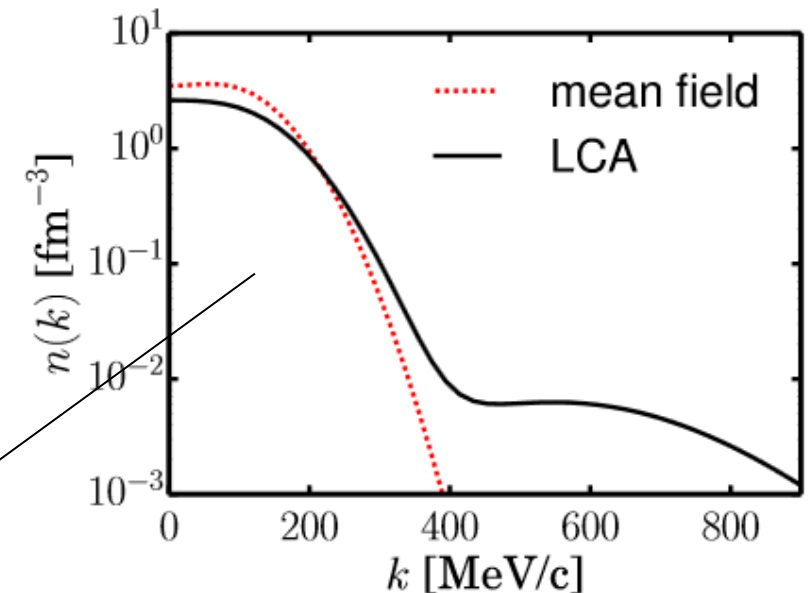
Quasi elastic versus multinucleon knockout processes : short range correlations



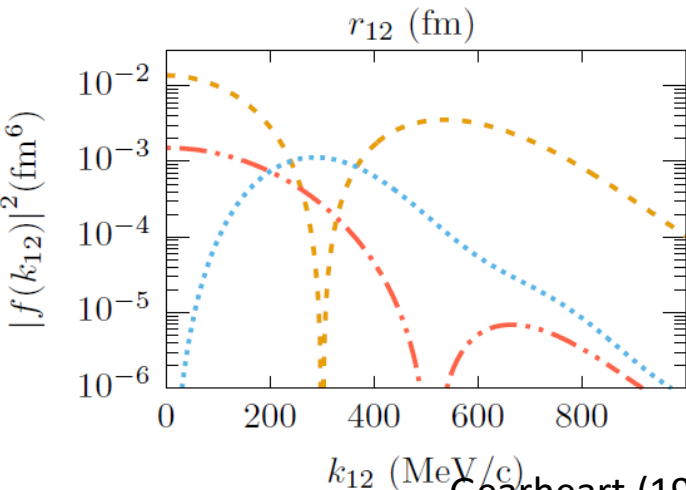
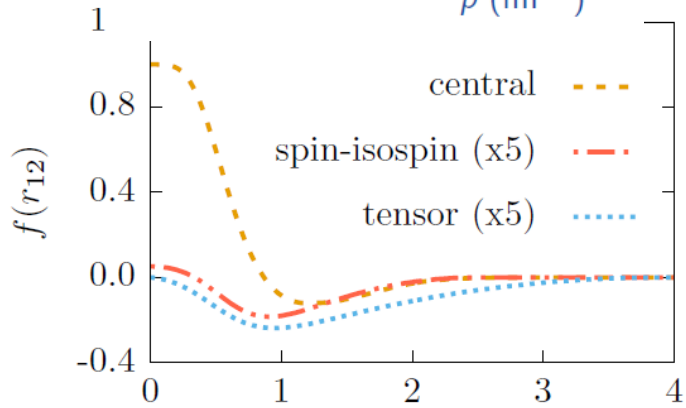
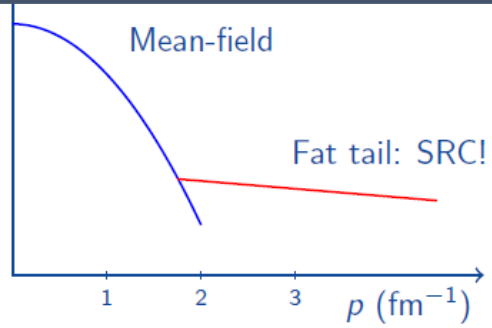
- SRC : short-range repulsive, tensor component of the nuclear force
- Individual nucleons receive large momenta compared to the Fermi momentum

IPM single-particle orbitals are depleted by SRC and higher energy levels are populated

- The short-range repulsive character of the nuclear force, which correlates with the Pauli exclusion principle, results in a large mean free path of the nucleons with respect to the size of the nucleus
- In an independent particle model nucleons move independently from each other in a mean field
- This approach fails to capture short-range features of nucleon-nucleon correlations



Quasi elastic versus multinucleon mechanisms : short range correlations



Gearheart (1994) Pieper (1992)

$$|\Psi\rangle = \frac{1}{\sqrt{\mathcal{N}}} \hat{\mathcal{G}} |\Phi\rangle \quad \text{with} \quad \hat{\mathcal{G}} \approx \hat{\mathcal{S}} \left(\prod_{i<j}^A [1 + \hat{l}(i, j)] \right)$$

Shifting the complexity induced by correlations from the wave functions to the operators

$$\hat{l}(i, j) = -g_c(r_{ij}) + f_{\sigma\tau}(r_{ij}) (\vec{\sigma}_i \cdot \vec{\sigma}_j) (\vec{\tau}_i \cdot \vec{\tau}_j) + f_{t\tau}(r_{ij}) \hat{S}_{ij} (\vec{\tau}_i \cdot \vec{\tau}_j),$$

$$\langle \Psi_f | \hat{J}_\mu^{\text{nuc}} | \Psi_i \rangle = \frac{1}{\sqrt{\mathcal{N}_i \mathcal{N}_f}} \langle \Phi_f | \hat{J}_\mu^{\text{eff}} | \Phi_i \rangle$$

$$\hat{J}_\mu^{\text{eff}} \approx \sum_{i=1}^A \hat{J}_\mu^{[1]}(i) + \sum_{i<j}^A \hat{J}_\mu^{[1],\text{in}}(i, j) + \left[\sum_{i<j}^A \hat{J}_\mu^{[1],\text{in}}(i, j) \right]^\dagger$$

$$\hat{J}_\mu^{[1],\text{in}}(i, j) = \left[\hat{J}_\mu^{[1]}(i) + \hat{J}_\mu^{[1]}(j) \right] \hat{l}(i, j)$$

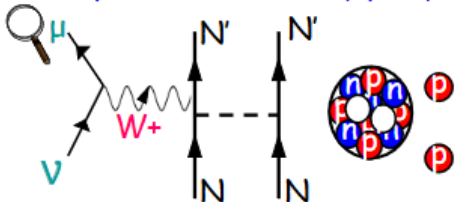
Quasi elastic versus multinucleon knockout processes : MECs

Genuine CCQE (1p-1h)



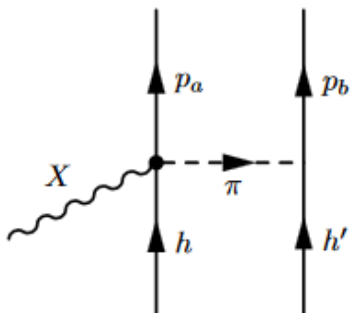
- QE processes are dominating the signal in experiments with average energies of a couple of hundreds of MeVs
- A thorough understanding of the QE cross section is extremely important as it is pivotal for energy reconstruction and oscillation analysis
- Correct identification of the reaction mechanism is important but not straightforward

Two particles-two holes (2p-2h)

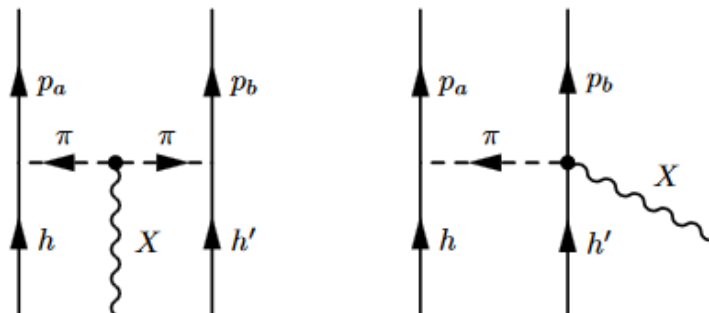


Meson-exchange currents

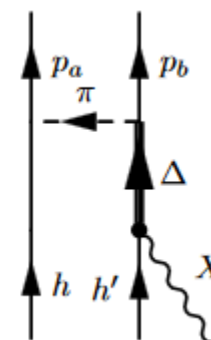
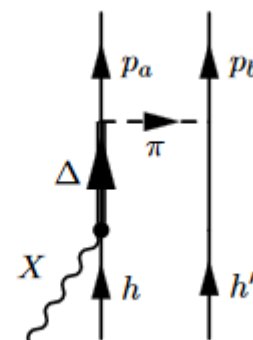
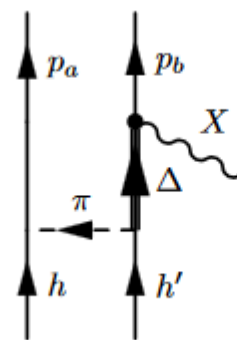
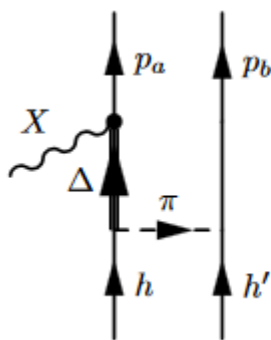
Seagull



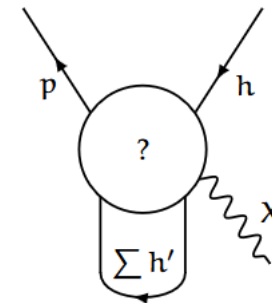
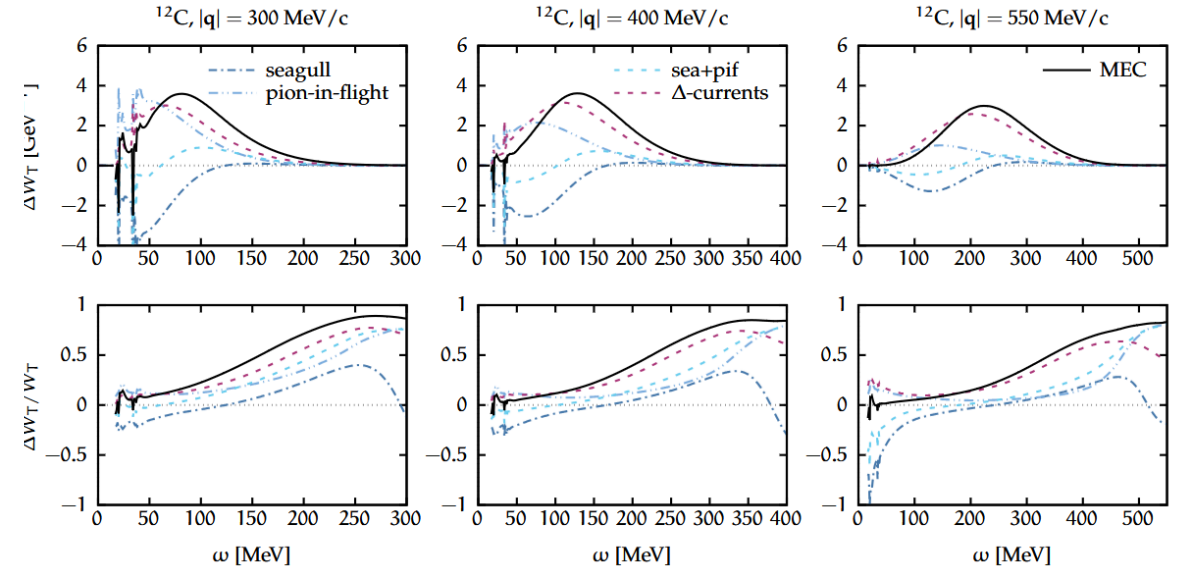
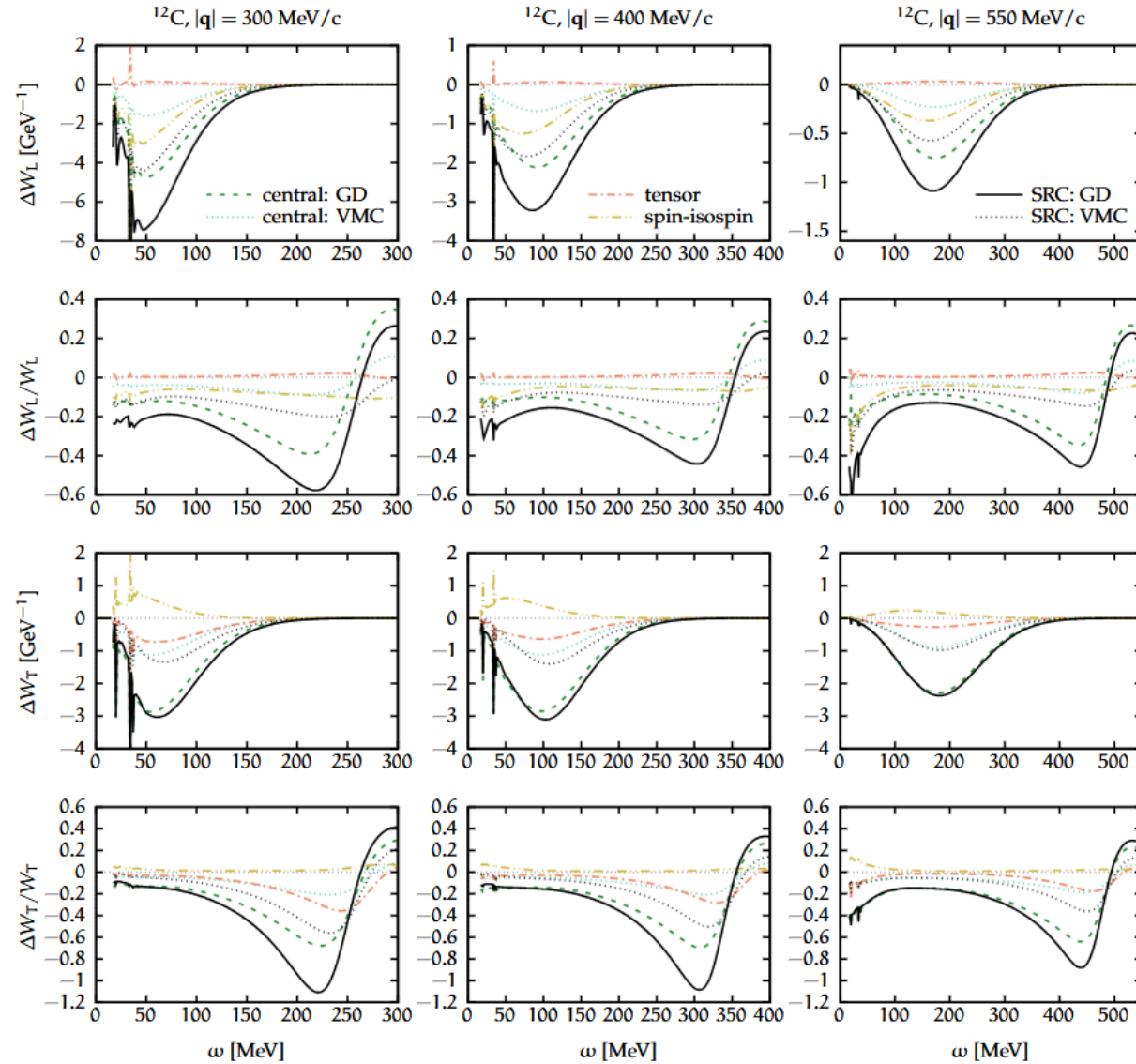
Pion in flight



Delta currents



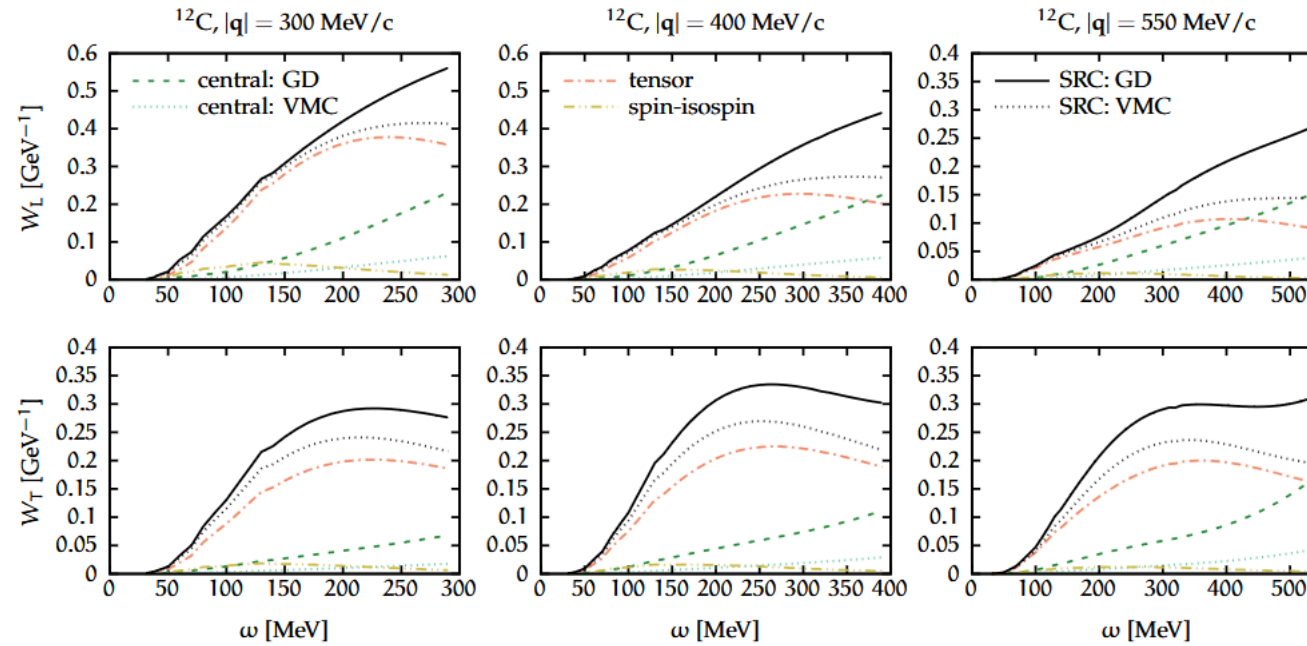
1p1h including SRC & MECs : e-scattering



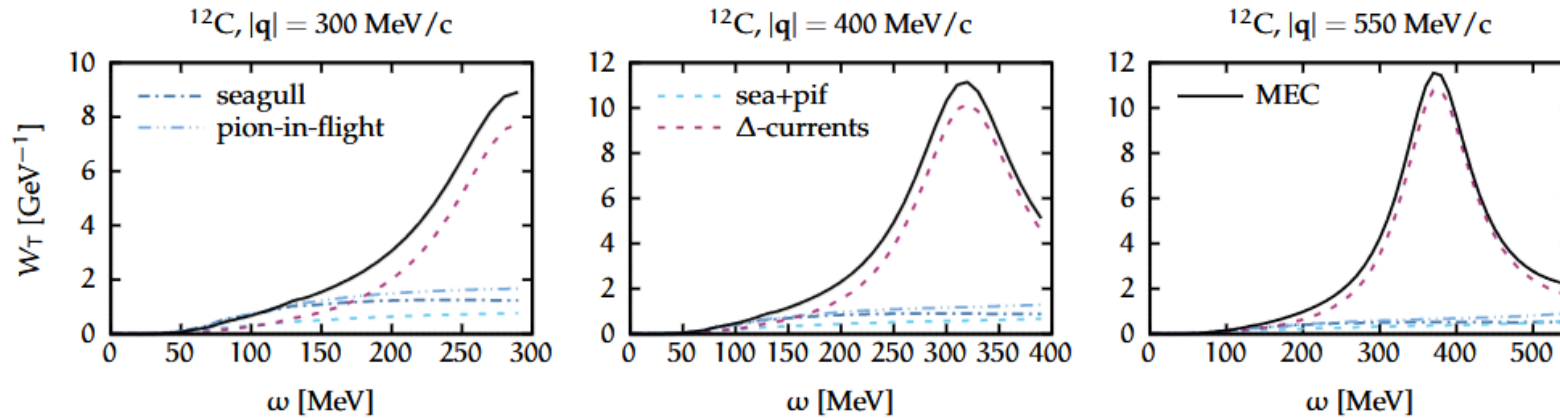
Kajetan Niewczas et al. arXiv2608.31044

2p2h e-scattering

SRC :

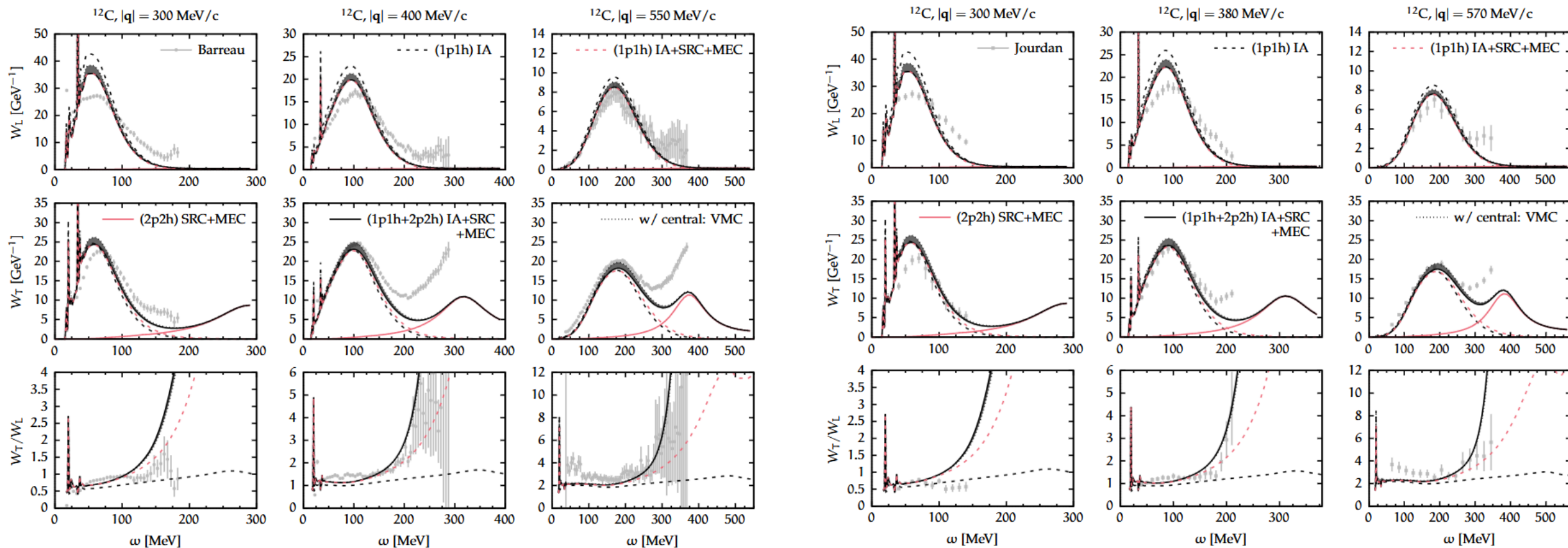


MECs :



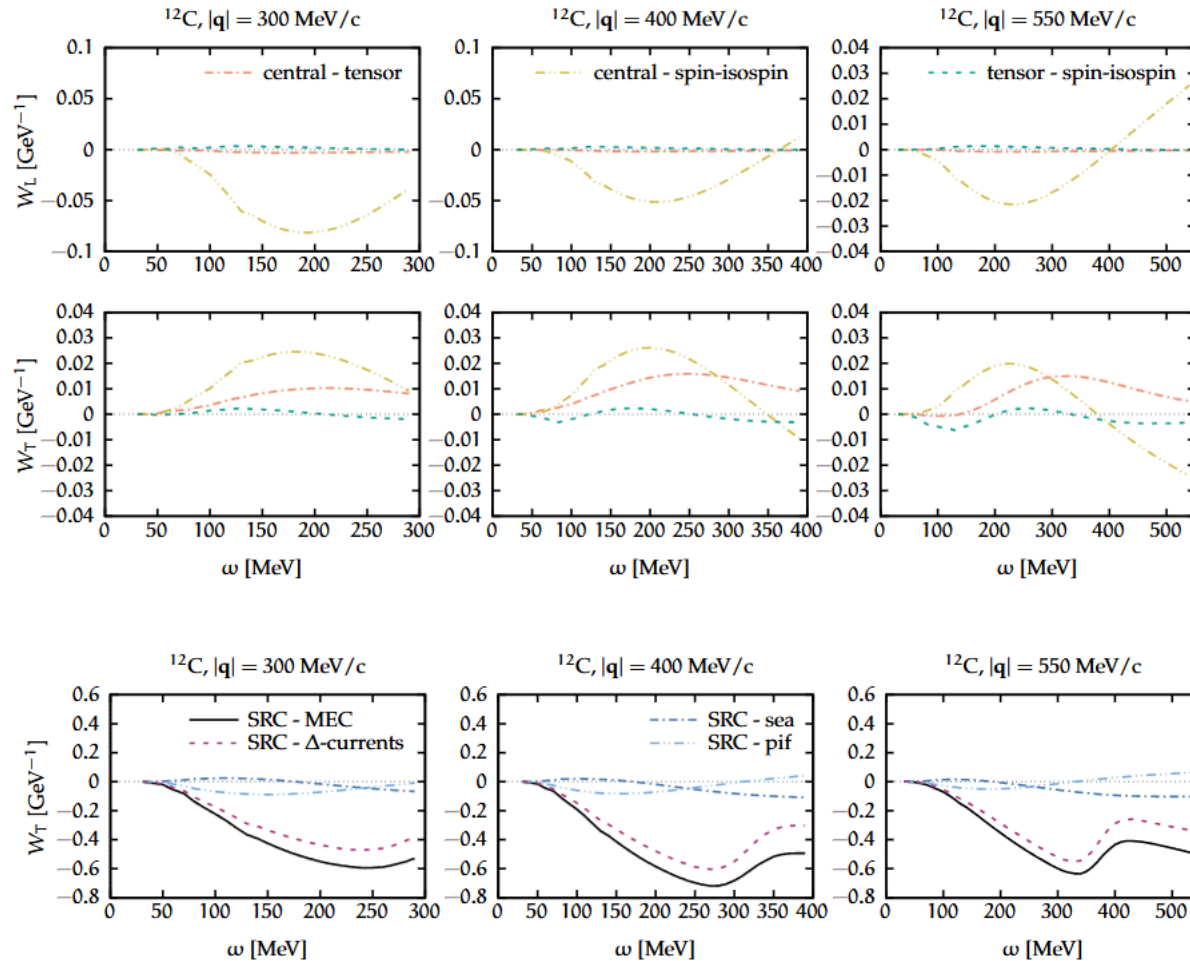
Kajetan Niewczas et al. arXiv2608.31044

1p1h+2p2h inclusive e-scattering



Kajetan Niewczas et al. arXiv2608.31044

2p2h inclusive e-scattering

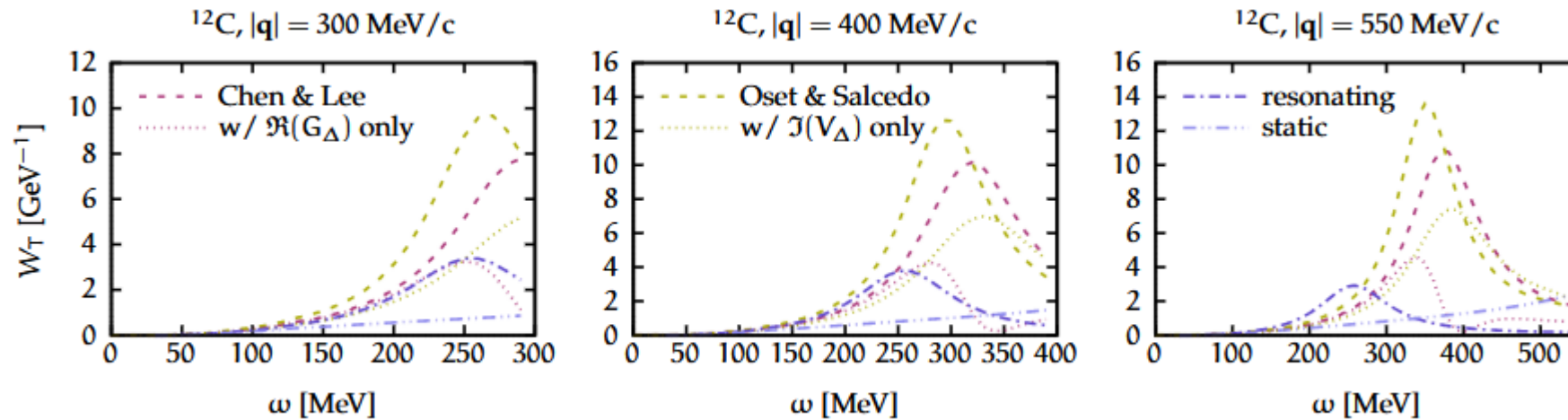


- Interference between SRCs and Δ MECs is substantial and destructive



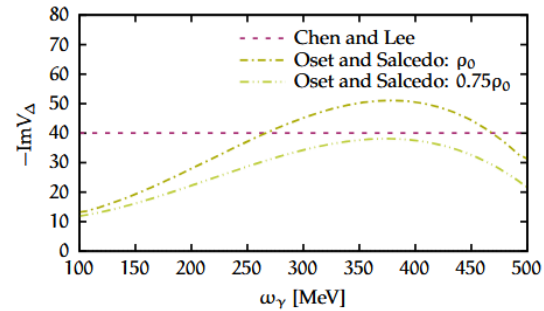
Kajetan Niewczas et al. arXiv2608.31044

2p2h inclusive e-scattering



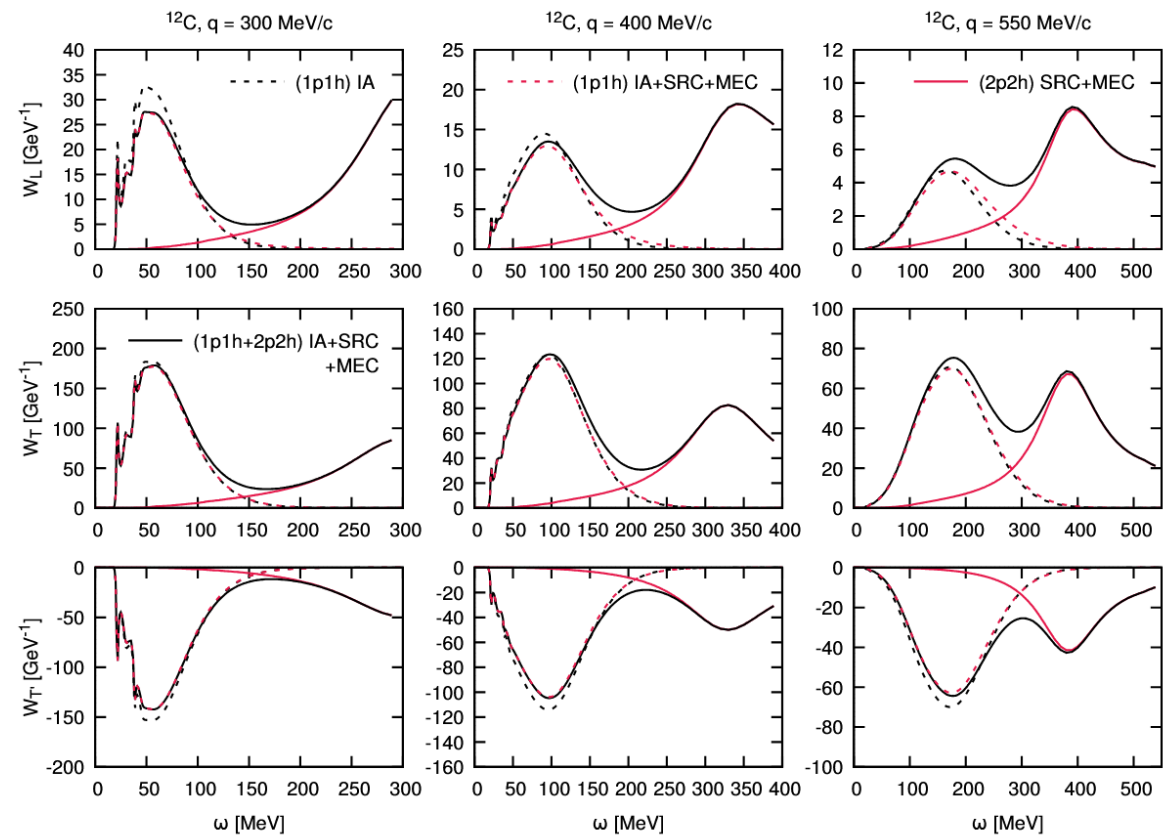
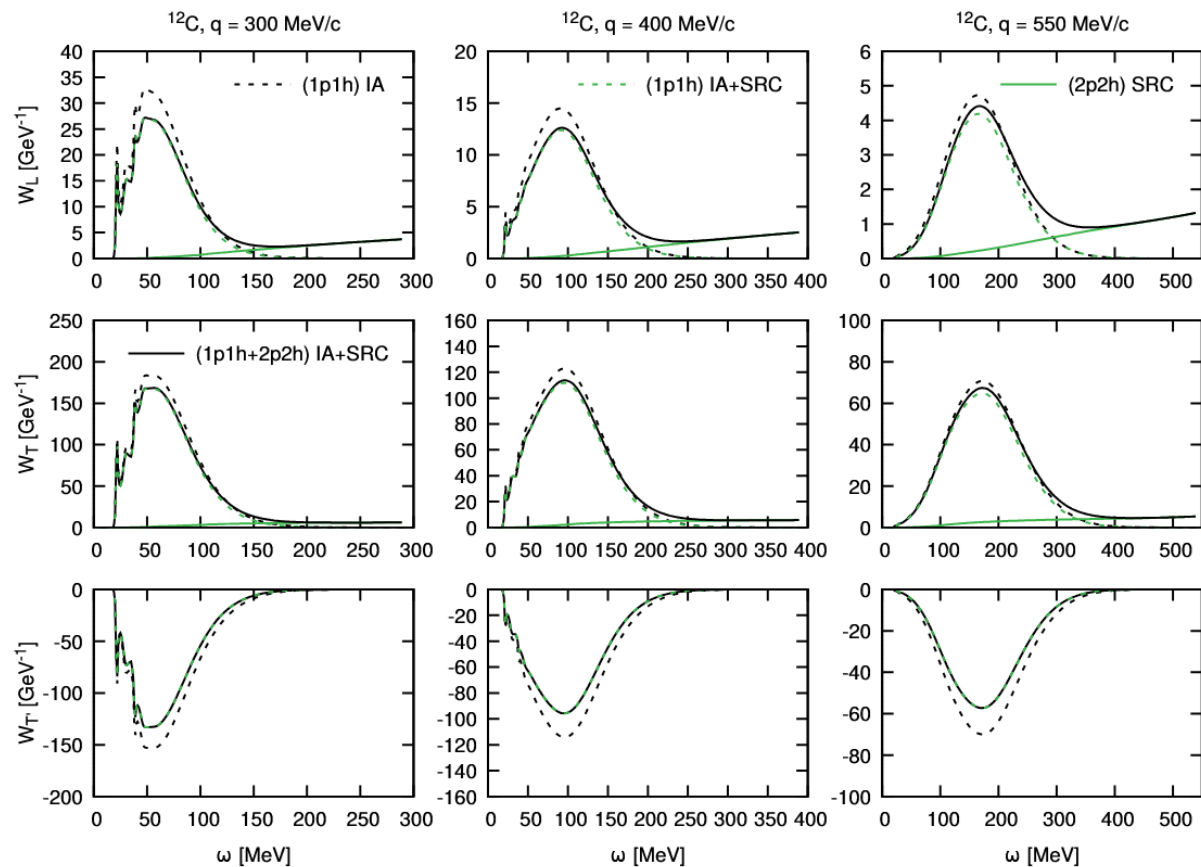
$$G_\Delta^{\text{res}} = \frac{1}{M_\Delta - \sqrt{s} - i/2\Gamma_\Delta^{\text{res}} + V_\Delta}$$

$$V_\Delta^{\text{Chen\&Lee}} = -30 - i40$$



Kajetan Niewczas et al. arXiv2608.31044

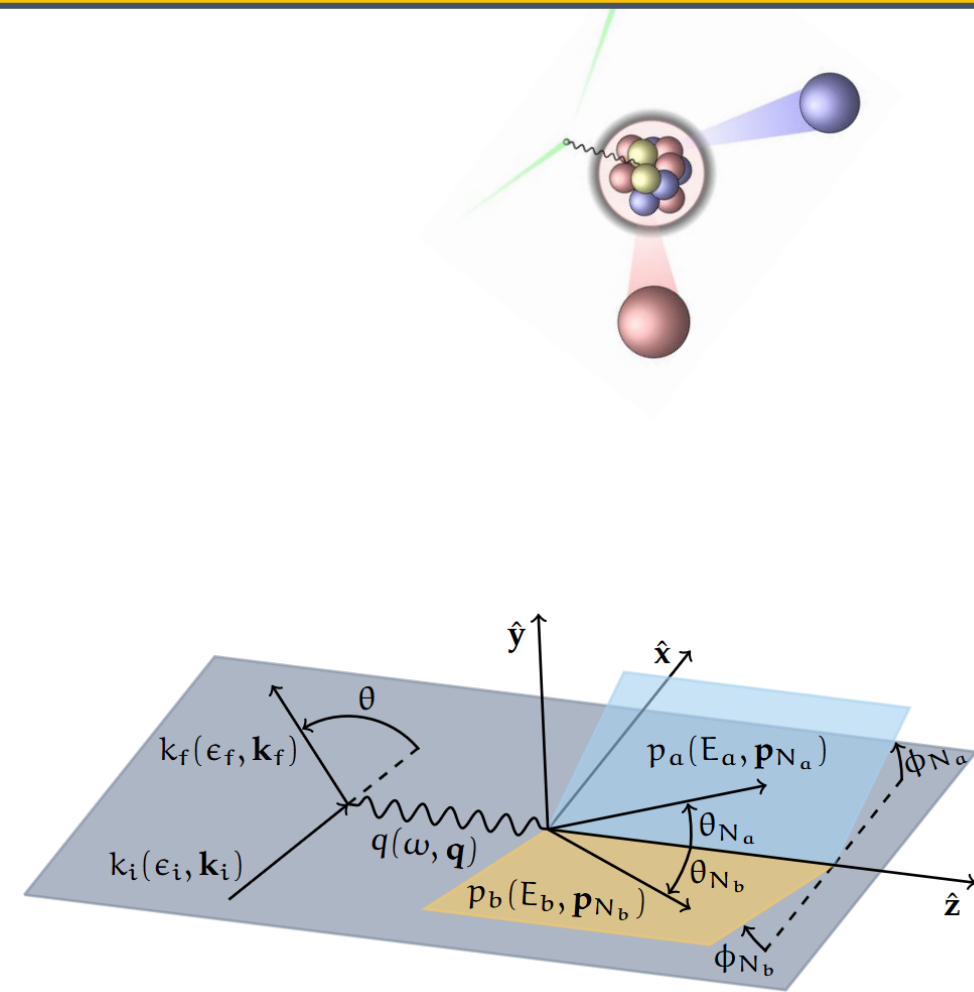
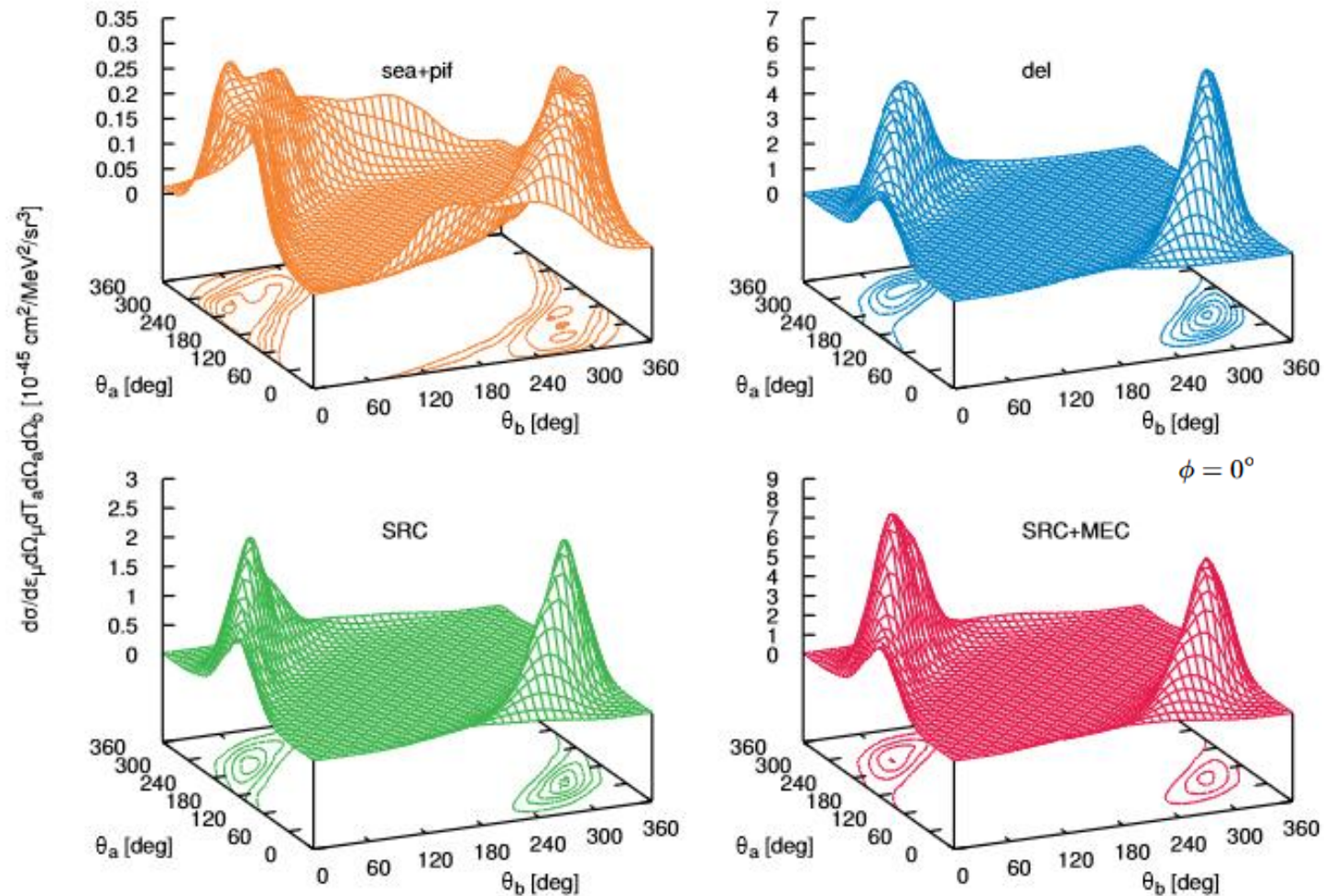
SRC & MECs : neutrinos



Kajetan Niewczas PhD Thesis 2023

2p2h cross sections : exclusive 2 nucleon knockout

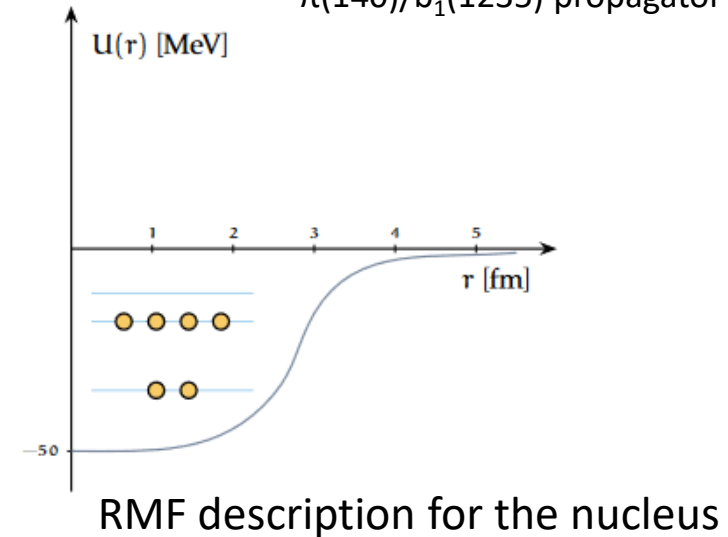
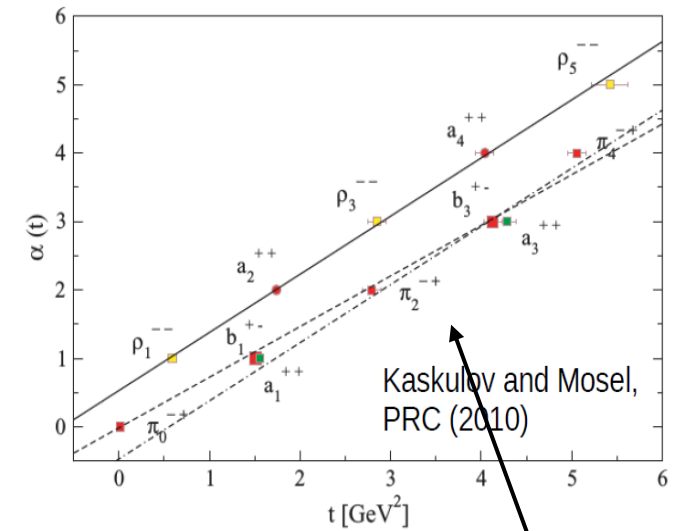
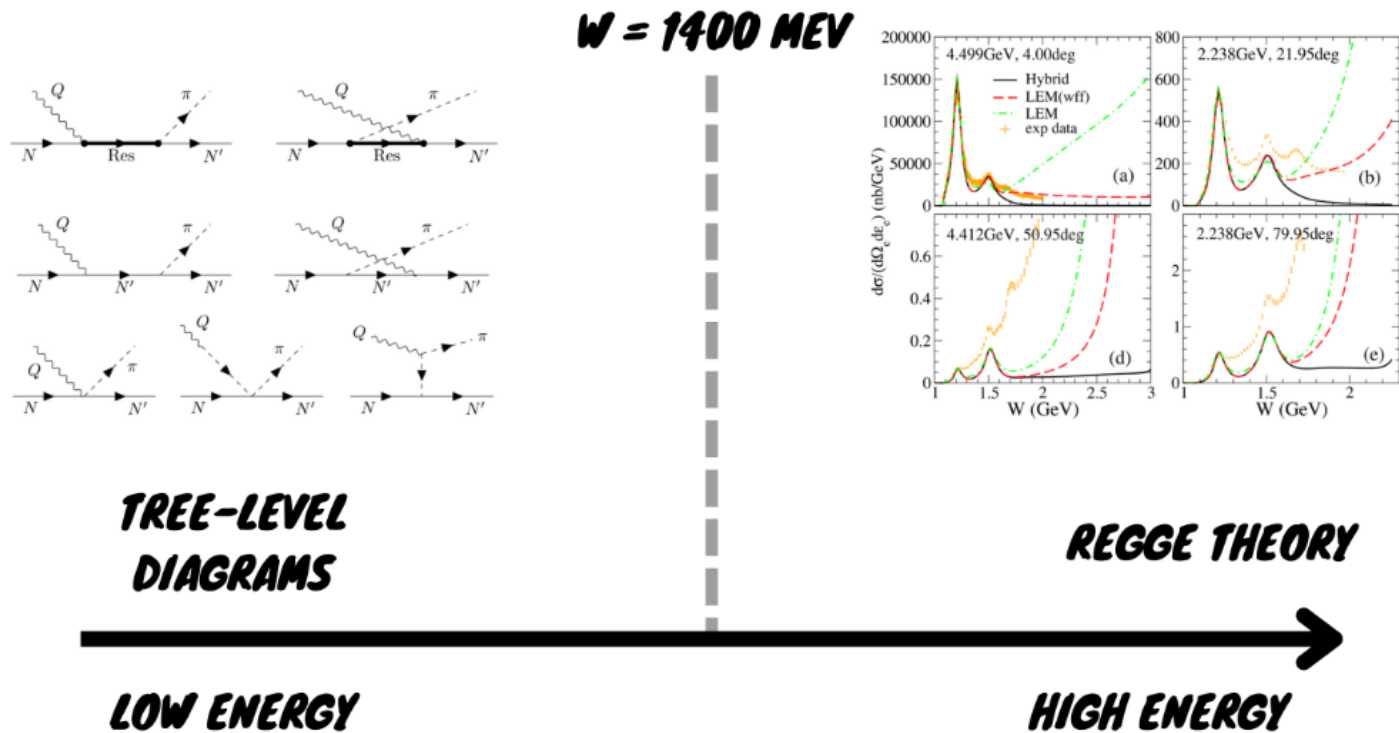
^{12}C , $\epsilon_{\nu\mu} = 750 \text{ MeV}$, $\epsilon_{\mu} = 550 \text{ MeV}$, $\theta_{\mu} = 15^\circ$



Kajetan Niewczas PhD Thesis 2023

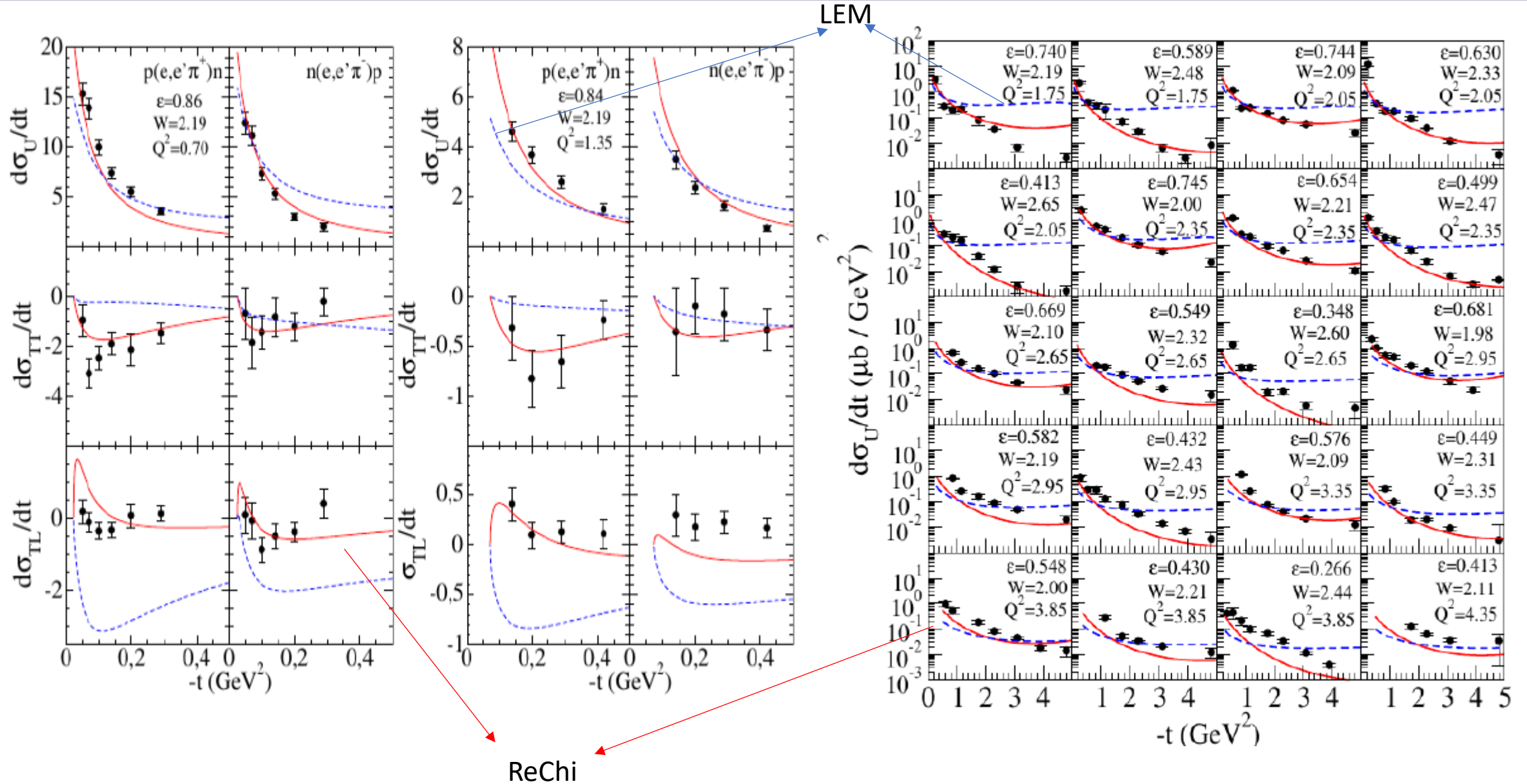
Single pion production - Ghent Hybrid model

- Designed aiming at the description of pion production off the nucleus
 - Low energy model : resonances + background from ChPT in HNV description
 - High energy model : Reggeized background to overcome shortcomings of the low energy description
- Aiming at fully exclusive description

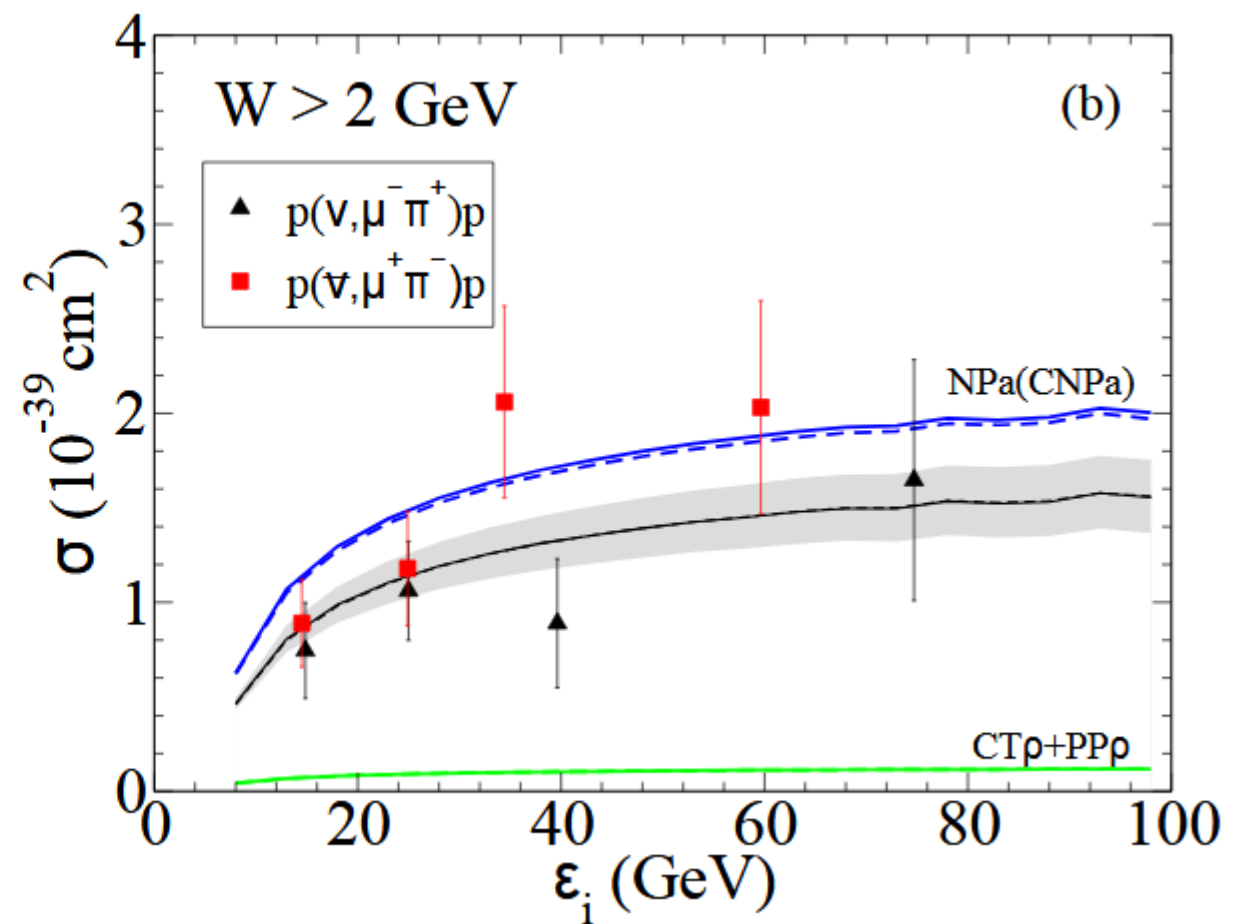
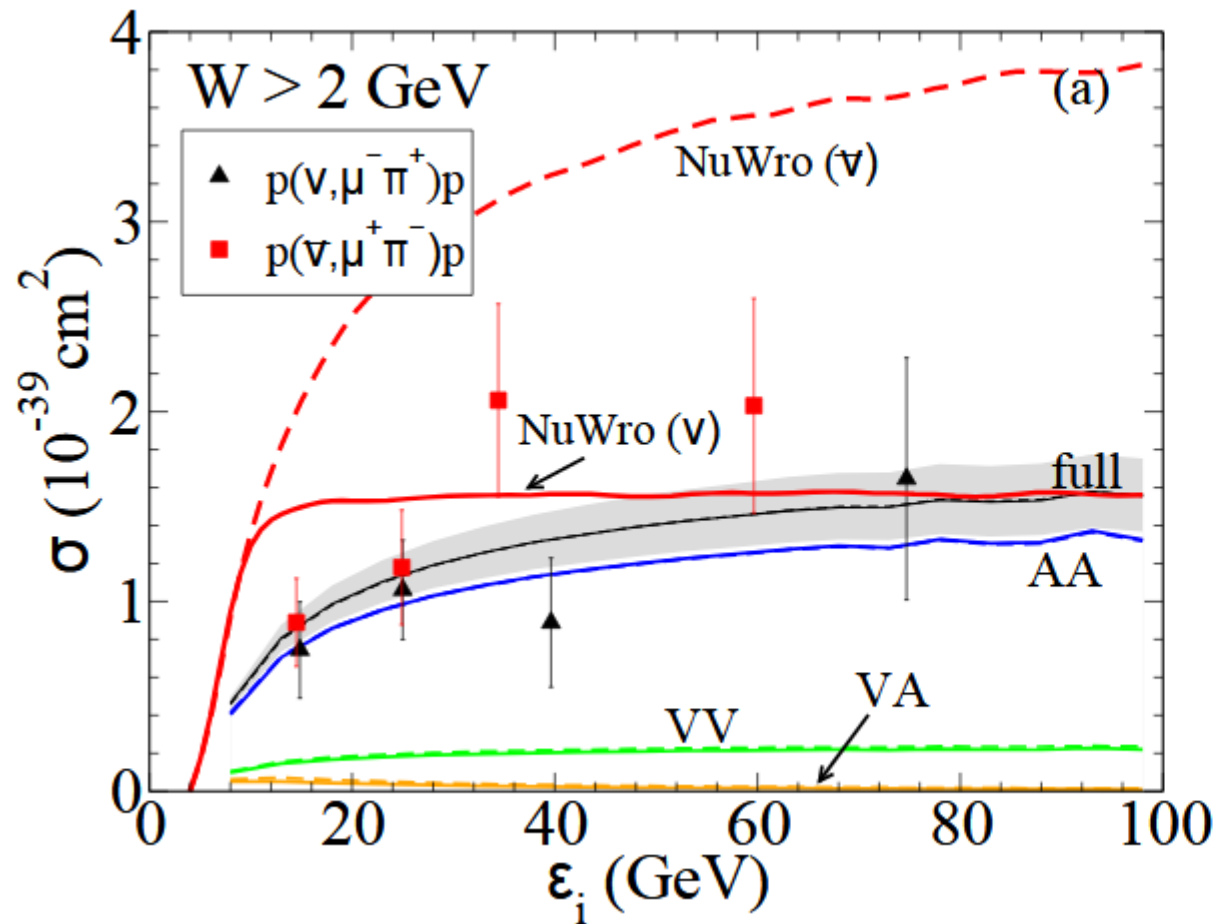


R. Gonzalez-Jimenez et al., Phys. Rev. D 97

Pion production : high energy model - electrons

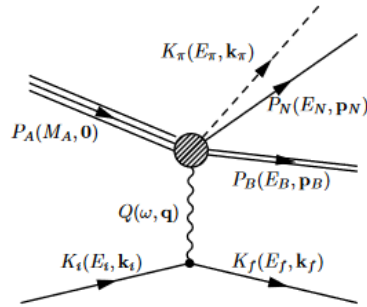


Pion production : high energy model - neutrinos



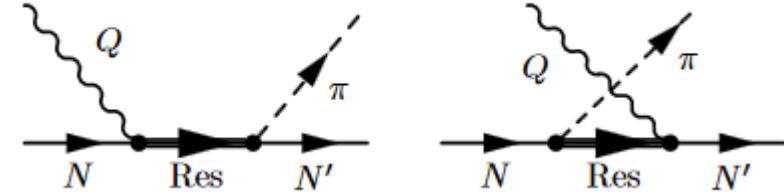
Data :P. Allen et al. Nucl. Phys. B264, 221 (1986).

Pion production : low energy model



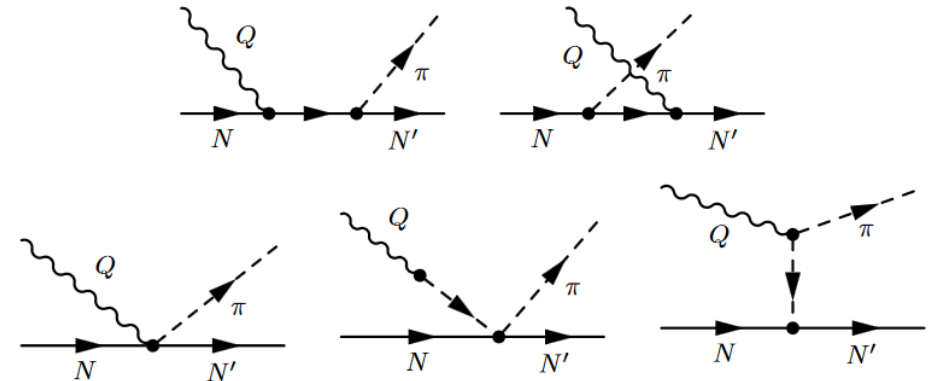
Resonances

P33 (1232), P11(1440), D13 (1520), S11 (1535)



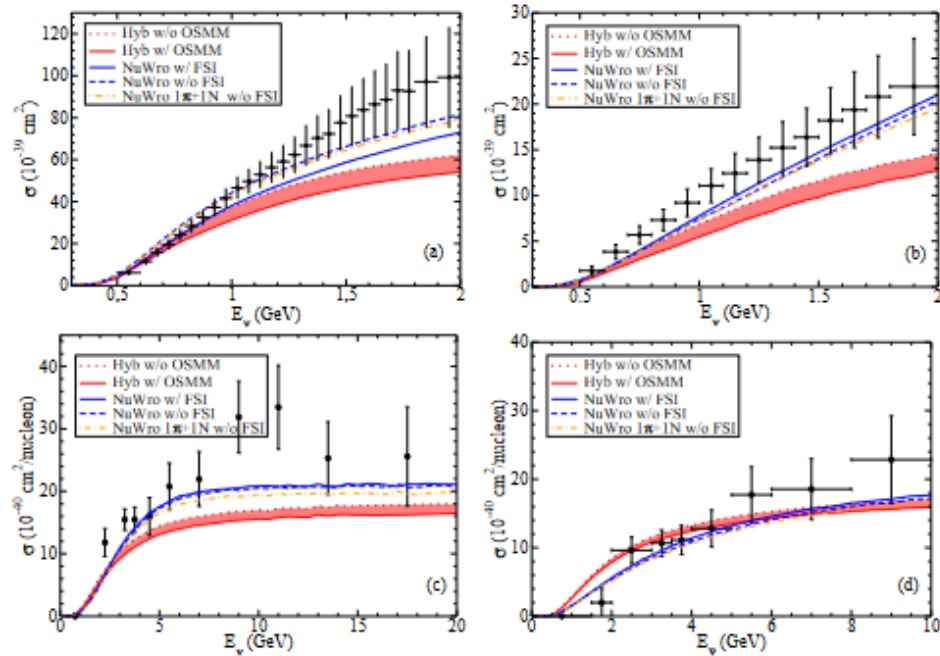
+

ChPT background



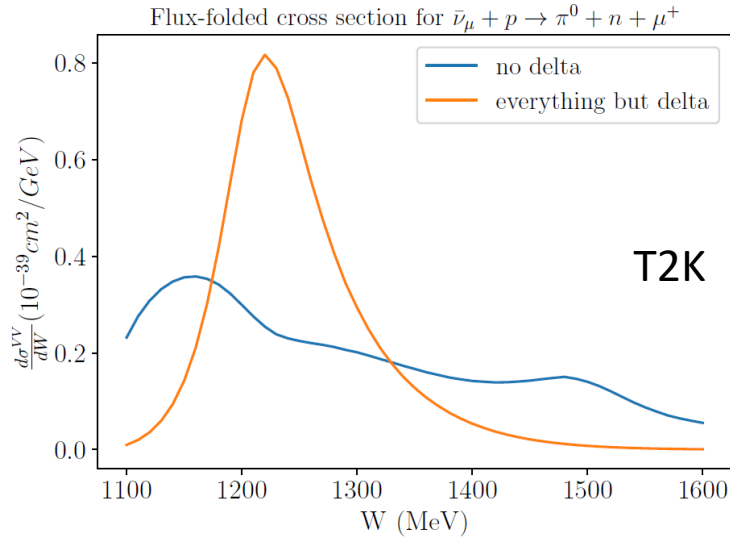
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Interferences

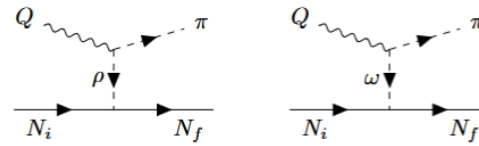


PRC 76, 033005 (2007), PRD87, 113009 (2013)

Recent developments single pion production : delta region



- Formulation of the formalism in terms of helicity amplitudes -> MAID form factors
- Multipole expansion -> Time-reversal invariance and unitarization of Delta and background contributions based on Watson's theorem
- Including ρ - and ω exchange diagrams



$$\tan \delta_{SPP} = \tan \delta_{\pi N}$$

$$i\pi \int dp p^2 T_{\pi\pi}(k_\pi, p; W) T_{\pi\gamma}^0(p, q; W) \times \delta(W - E_\pi(p) - E_N(p)) \times \frac{k_\pi \cdot E_N}{E_\pi} T_{\pi\pi}(k_\pi, q; W) = i\pi k_\pi^2 \frac{E_N}{W} T_{\pi\pi}(k_\pi, k_\pi; W) T_{\pi\gamma}^0(k_\pi, q; W) - i p_\pi T_{\pi\pi}(k_\pi, k_\pi; W) T_{\pi\gamma}^0(k_\pi, q; W).$$

$$\pi^- p \langle \lambda_F | T^{EM} | \lambda_I \rangle_{\gamma n} = \frac{\sqrt{2}}{3} B_{3/2} - \frac{\sqrt{2}}{3} B_{1/2} - \sqrt{\frac{2}{3}} B_{1/2}^S$$

$$\begin{aligned} \left\langle \frac{1}{2} \middle| T_{r=-1} \middle| \frac{3}{2} \right\rangle &= \chi_{f+}^{\dagger} (\epsilon_\mu^{r=-1} \mathcal{F}^\mu) \chi_{i+} = \frac{i}{\sqrt{2}} \sin \theta_\pi \cos \frac{\theta_\pi}{2} (F_3 + F_4), \\ \left\langle -\frac{1}{2} \middle| T_{r=-1} \middle| \frac{3}{2} \right\rangle &= \chi_{f-}^{\dagger} (\epsilon_\mu^{r=-1} \mathcal{F}^\mu) \chi_{i+} = \frac{i}{\sqrt{2}} \sin \theta_\pi \sin \frac{\theta_\pi}{2} (-F_3 + F_4), \\ \left\langle \frac{1}{2} \middle| T_{r=-1} \middle| \frac{1}{2} \right\rangle &= \chi_{f+}^{\dagger} (\epsilon_\mu^{r=-1} \mathcal{F}^\mu) \chi_{i-} = \frac{i}{\sqrt{2}} \left(2 \cos \frac{\theta_\pi}{2} (F_1 - F_2) - \sin \theta_\pi \sin \frac{\theta_\pi}{2} (F_3 - F_4) \right), \\ \left\langle -\frac{1}{2} \middle| T_{r=-1} \middle| \frac{1}{2} \right\rangle &= \chi_{f-}^{\dagger} (\epsilon_\mu^{r=-1} \mathcal{F}^\mu) \chi_{i-} = \frac{i}{\sqrt{2}} \left(-2 \sin \frac{\theta_\pi}{2} (F_1 + F_2) - \sin \theta_\pi \cos \frac{\theta_\pi}{2} (F_3 + F_4) \right). \end{aligned}$$

$$\pi^0 n \langle \lambda_F | T^{EM} | \lambda_I \rangle_{\gamma n} = \frac{\sqrt{2l+3}}{3} B_{3/2} + \frac{1}{3} B_{1/2} + \sqrt{\frac{1}{3}} B_{1/2}^S.$$

Appendix A: Definitions of helicity spinors

Appendix B: Total angular momentum states

Appendix C: Unitarity and angular momentum states

Appendix D: Isospin separation of multipoles

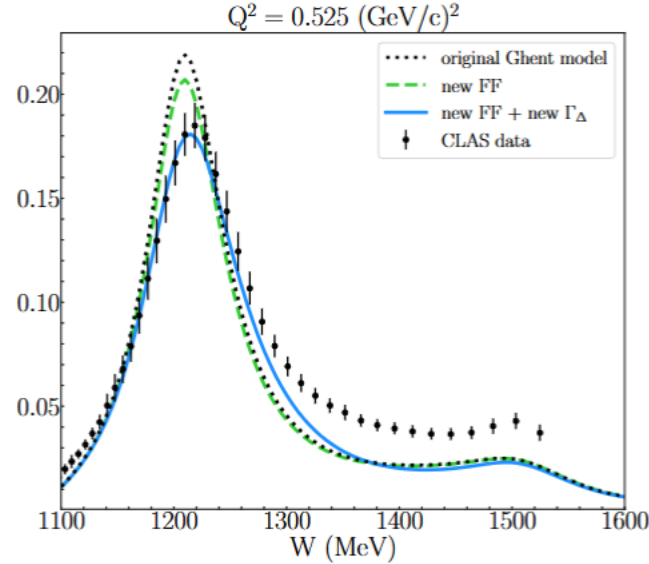
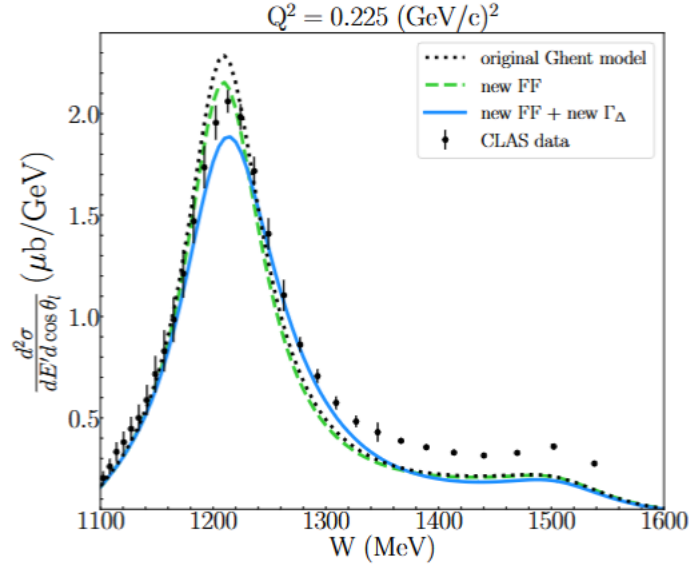
Appendix E: From helicity to multipole amplitudes

Appendix F: Responses in terms of multipoles

Appendix G: K- and T-matrix of the background contributions

M. Hoof et al, arXiv:2603.29486

Recent developments single pion production

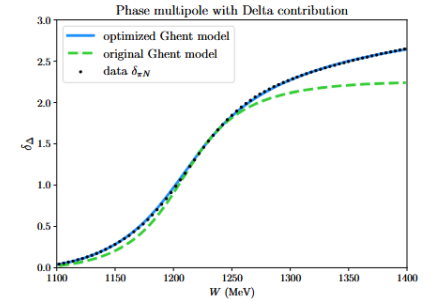


Delta :

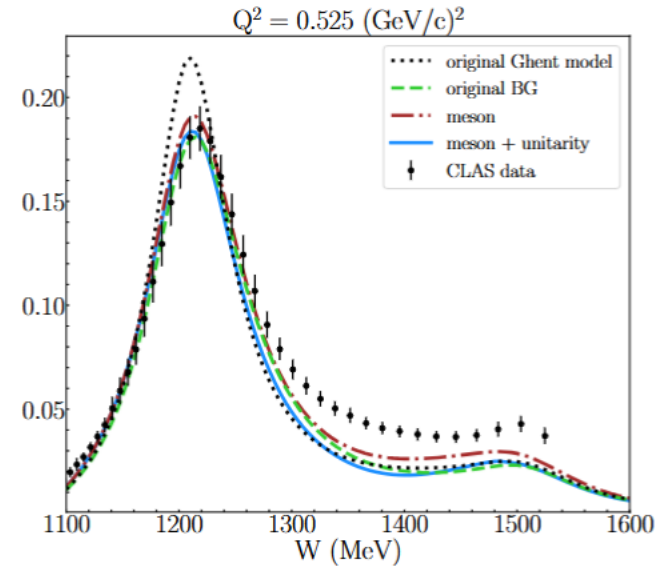
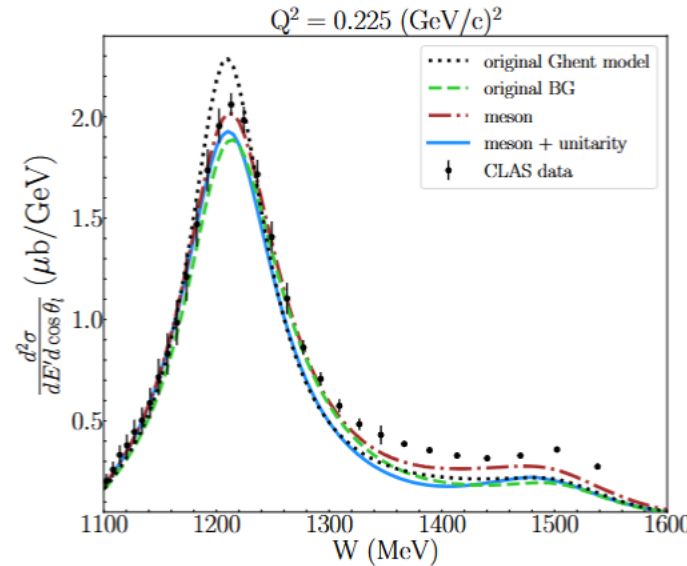
- New form factors
- Unitarization of Delta by adjusting the width to comply with Watson's theorem



$$\Gamma_{\Delta} = \frac{M_{\Delta}^2 - W^2}{M_{\Delta}} \tan(\delta_{\pi N}(W))$$



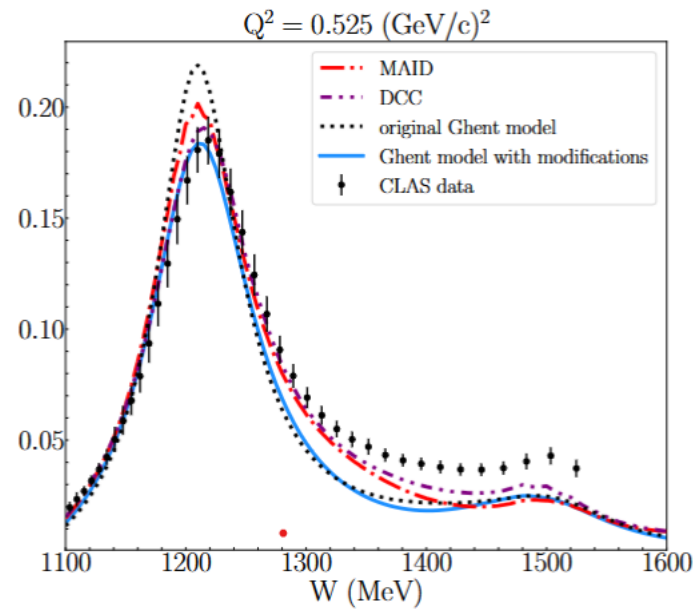
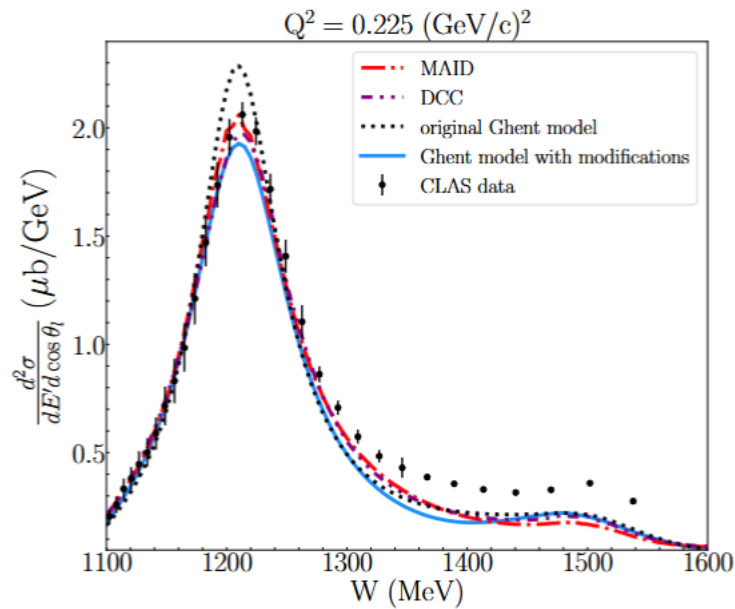
- Unitarization of the background using K-matrix theory
- ρ - and ω exchange diagrams



M. Hooft et al, arXiv:2603.29486

Recent developments single pion production

- Unitarization of Delta and background contributions using Watson's theorem : multipole expansion
- New form factors (MAID) : formulation of model in terms of helicity amplitudes
- Including ρ - and ω exchange diagrams

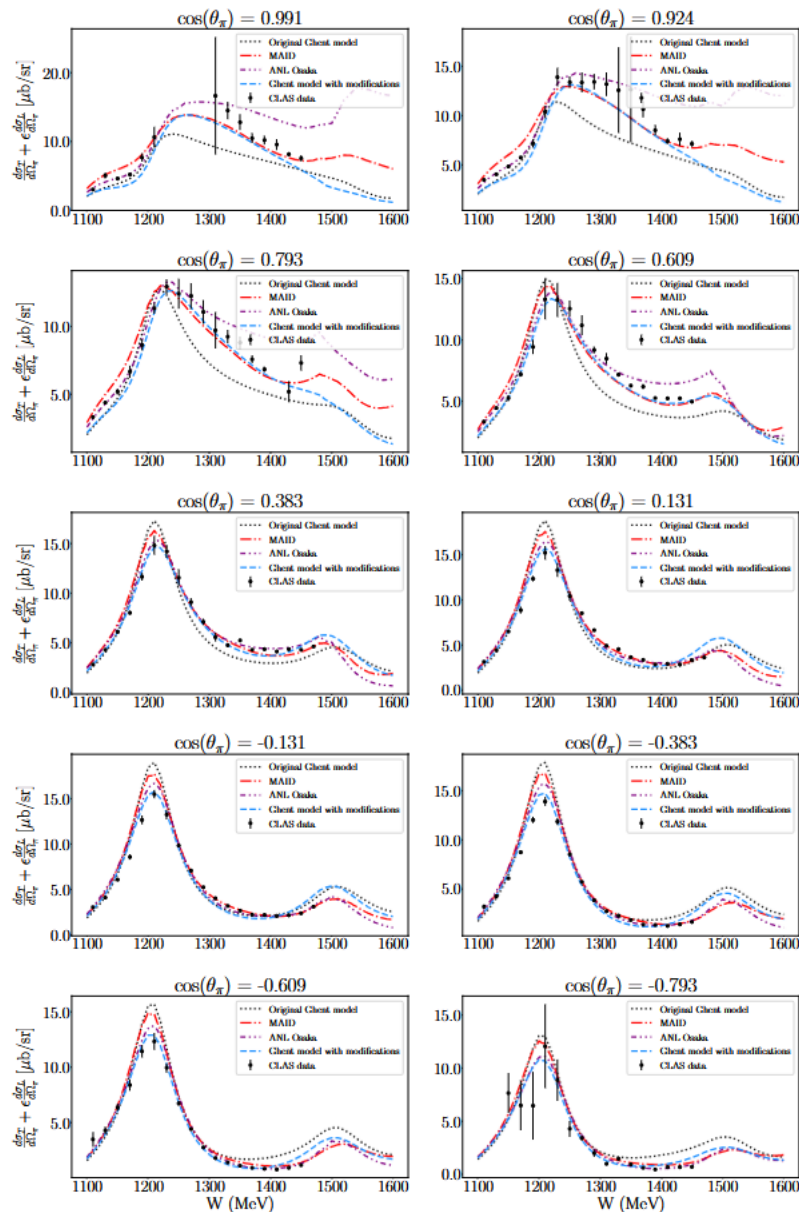


Comparison with MAID, Osaka DCC and CLAS data

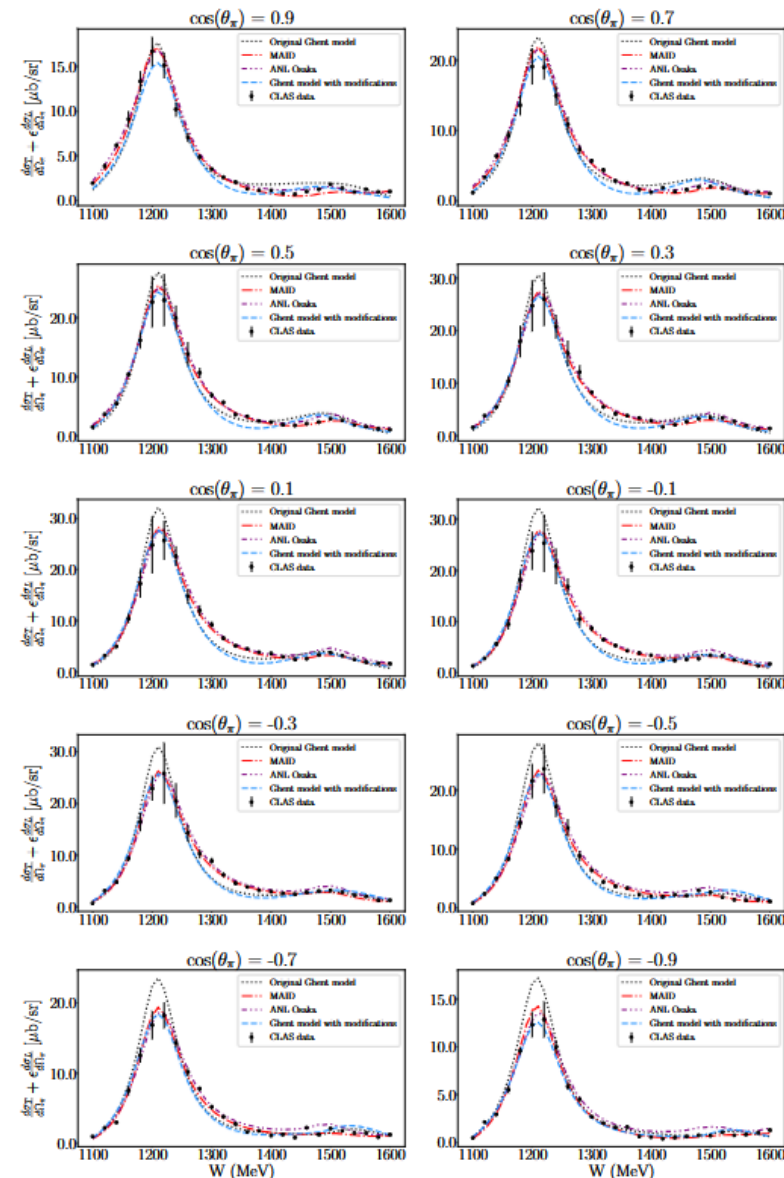
Recent developments single pion production

$Q^2 = 0.4 \text{ (GeV/c)}^2$

π^+



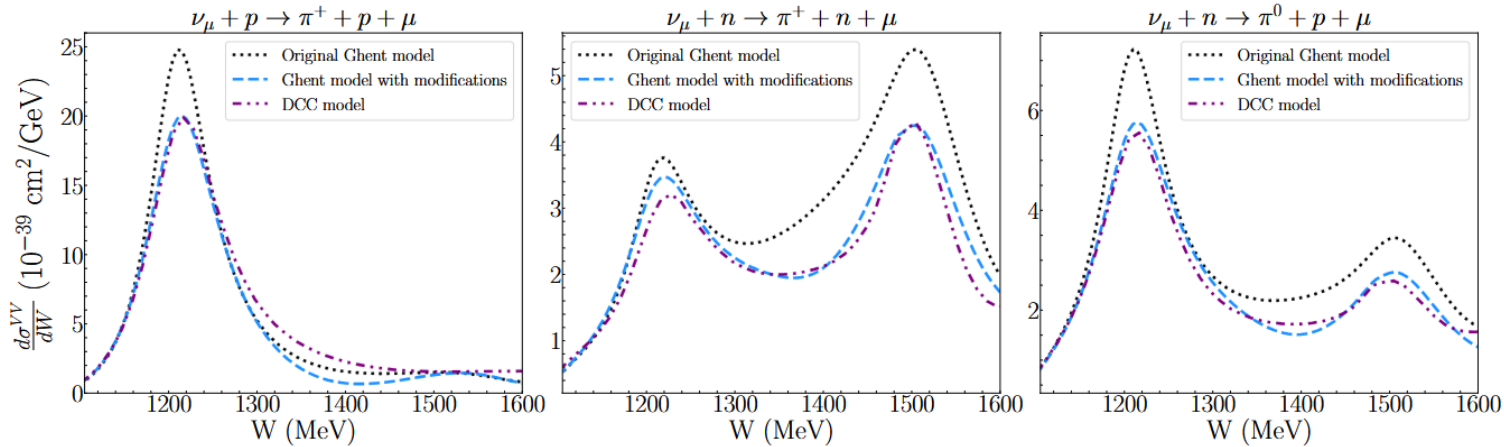
π^0



Modeling single pion production

Work in progress :

- Add axial contributions

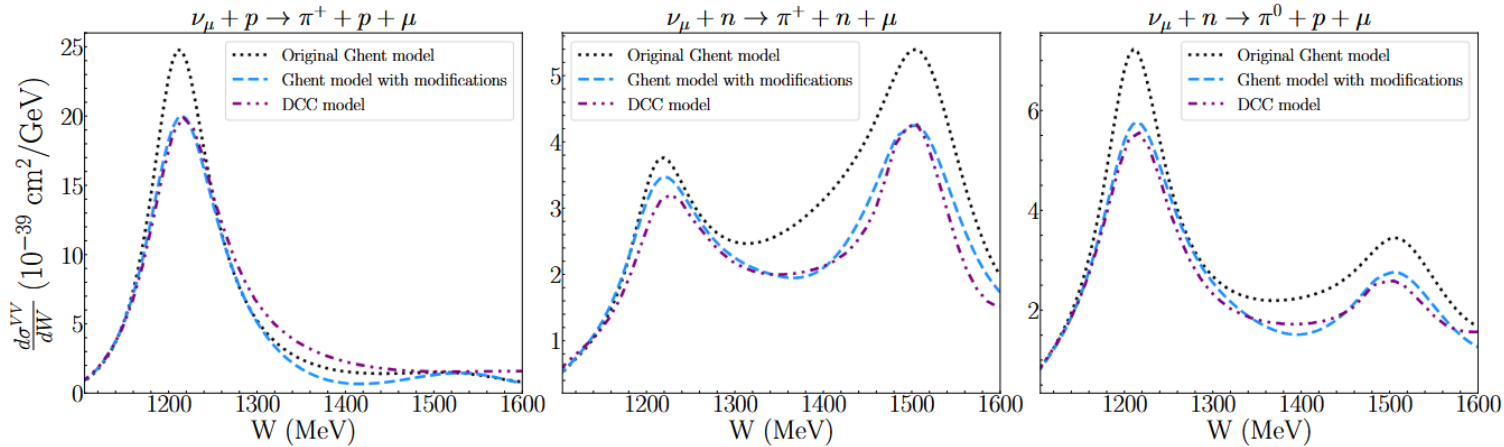


BEBC flux : vector contribution

Modeling single pion production

Work in progress :

- Add axial contributions
- Add higher resonances



BEBC flux : vector contribution

Modeling single pion production

Work in progress :

- Add axial contributions
- Constituent couplings : remove spurious contributions

Modeling single pion production

Work in progress :

- Add axial contributions
- Constituent couplings : remove spurious contributions
- Embed in nuclear framework

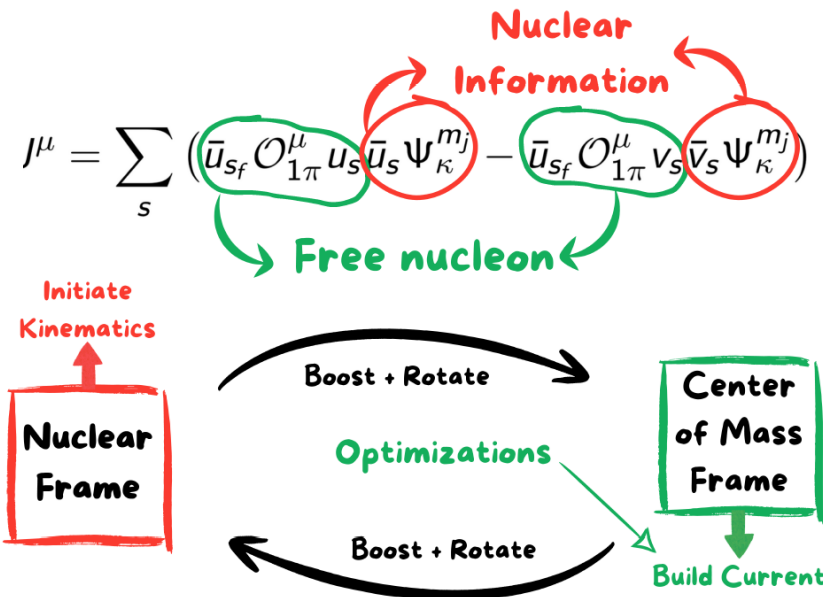
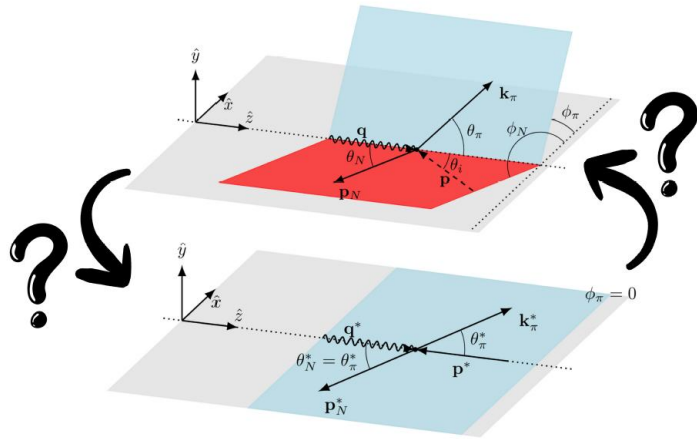
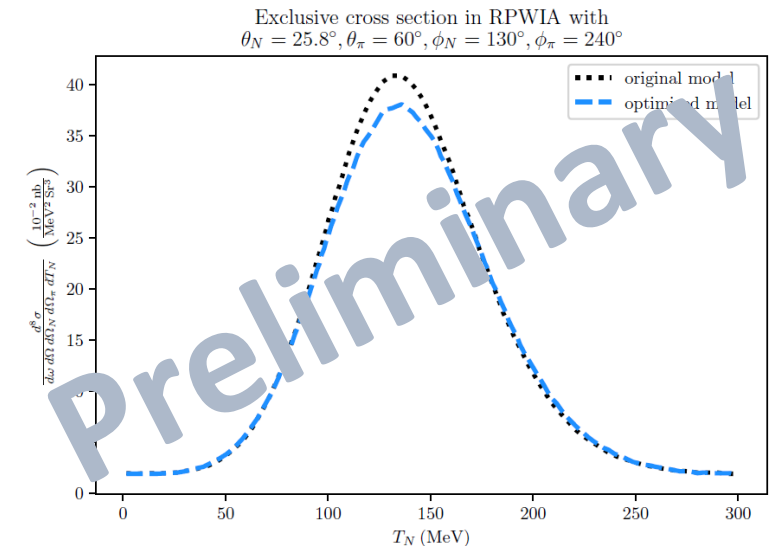


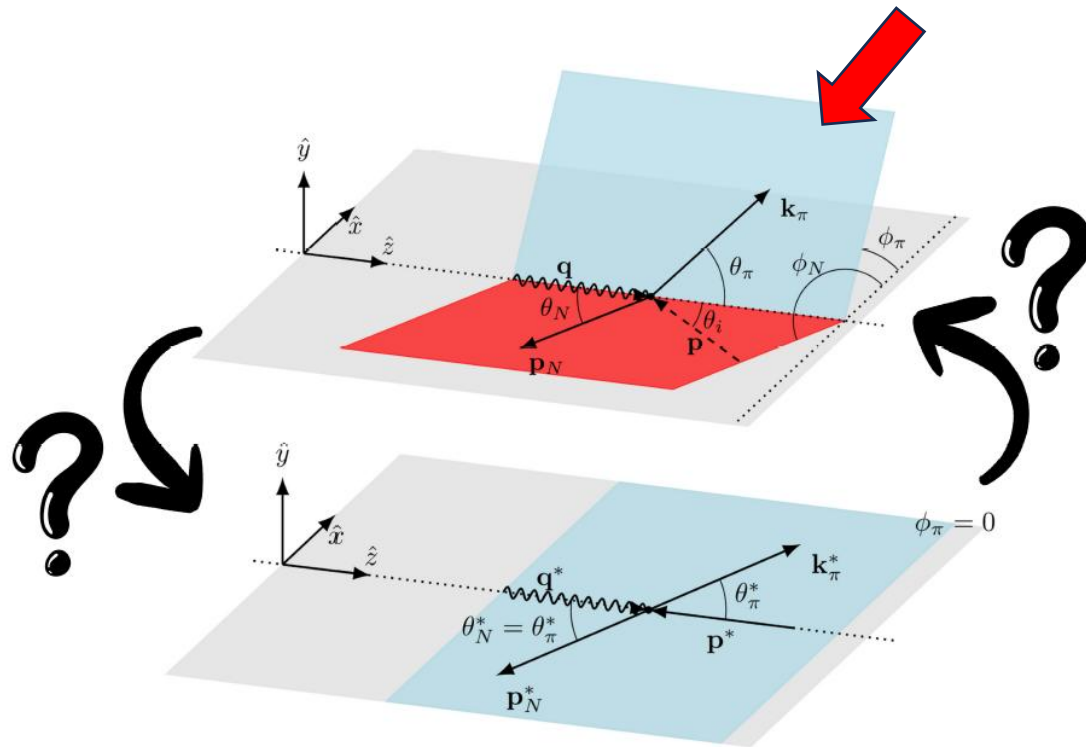
Figure : M. Hooft



Modeling single pion production

Work in progress :

- Add axial contributions
- Constituent couplings : remove spurious contributions
- Embed in nuclear framework
- Distortion of the pion



Ghent models in NuWro : SPP

- Default NuWro

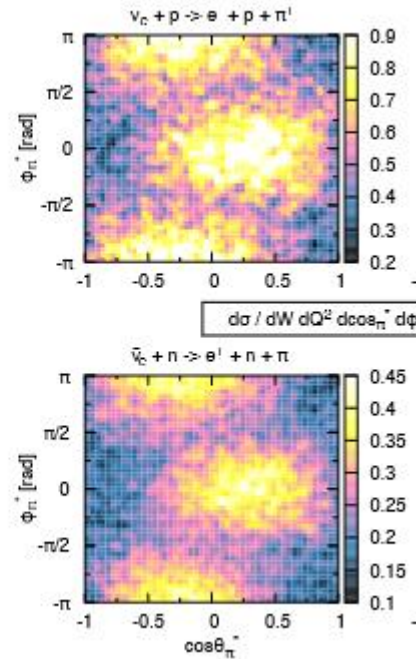
- Free nucleon

- Fixed kinematics:

$$E = 1 \text{ GeV}$$

$$Q^2 = 0.1 \text{ GeV}^2$$

$$W = 1230 \text{ MeV}$$



- Ghent LEM

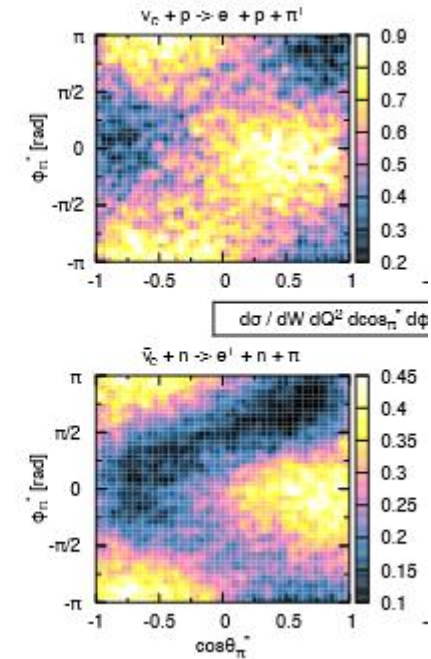
- Free nucleon

- Fixed kinematics:

$$E = 1 \text{ GeV}$$

$$Q^2 = 0.1 \text{ GeV}^2$$

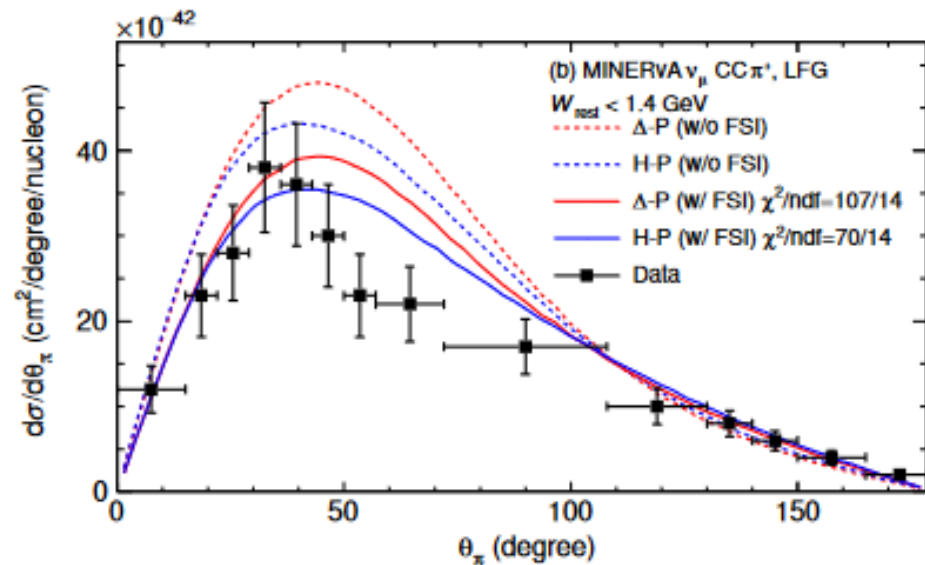
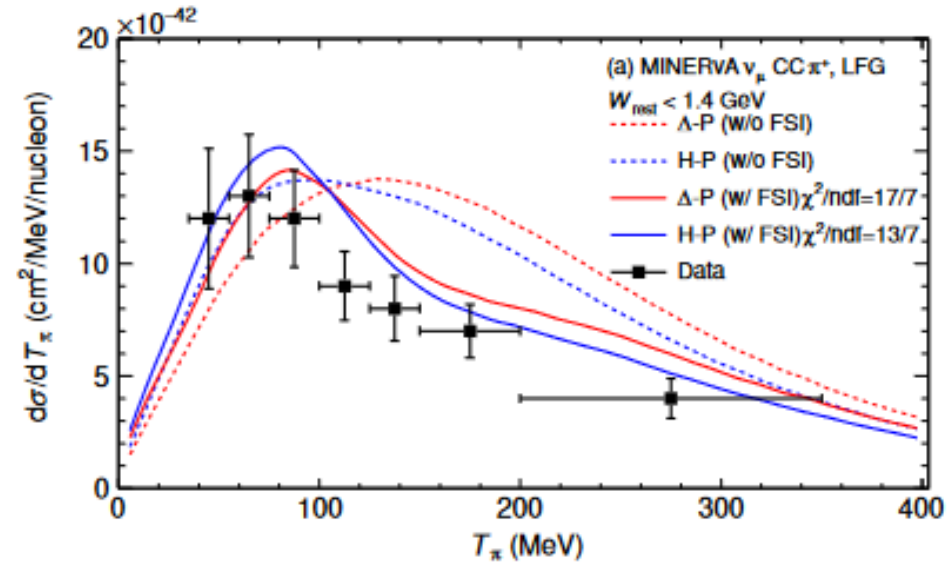
$$W = 1230 \text{ MeV}$$



K. Niewczas et al. PRD103, 053003 (2021)

Need to go beyond isotropic approaches in MC generators and include detailed microscopic calculation results in the simulation

Ghent model in NuWro



Integration of Ghent hybrid model in NuWro for pion production off the nucleus

Q. Yan et al. arXiv 2405.05212